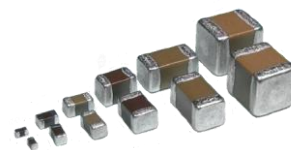


Multilayer Ceramic Chip Capacitors

How to Order

■ Features

- Kyocera's series of Multilayer Ceramic Chip Capacitors are designed to meet a wide variety of needs. We offer a complete range of products for both general and specialized applications.
- We have a network worldwide in order to supply our global customer bases quickly and efficiently.
- All our products are highly reliable due to their monolithic structure of high-purity and superfine uniform ceramics and their integral internal electrodes.
- Our stringent quality control in every phase of production from material procurement to shipping ensures consistent manufacturing and superior quality.
- Kyocera components are available in a wide choice of dimensions, temperature characteristics, rated voltages, and terminations to meet specific configurational requirements.



e.g.)

KGM 03 C R5 0J 225 M H □ □ □ □

① ② ③ ④ ⑤ ⑥ ⑦ ⑧ Option Code (When needed)

*Note: "I" (capital i) and "O" (capital o) are omitted from part numbers to avoid confusion with numbers "1" and "0".

- ① Series : KGM Series(General)
- ② Size (EIA) : 0201
- ③ Thickness (max.) : 0.39mm
- ④ Dielectric : Operating Temperature Range: -55 to 85°C/
ΔC max.: ±15%/ Standard Temperature: 25°C/
6.3Vdc
- ⑤ Rated Voltage : 2.2μF
- ⑥ Capacitance : ±20%
- ⑦ Tolerance : Taping Material Paper/ Taping Width 8mm/
Cavity Pitch 2mm/ Reel Size φ180
- ⑧ Packaging

① Series Code

Code	Type
KGM	General
KGT	Low Profile
KGU	High-Q
KAM	Automotive
KGN	Three Terminal Capacitors

② Size Code

Code	EIA	JIS
02	01005	0402
03	0201	0603
05	0402	1005
15	0603	1608
21	0805	2012
31	1206	3216
32	1210	3225

③ Thickness(max.)

Thickness Code	Code	EIA	Thickness (mm max.)
A	02	01005	0.22
	03	0201	0.33
	05	0402	0.55
	15	0603	0.9
	21	0805	1.45
	31	1206	1.8
	32	1210	2.7
B	03	0201	0.35
	05	0402	0.65
C	03	0201	0.39
	05	0402	0.7
	15	0603	1.0
D	03	0201	0.55
	05	0402	0.8
F	31	1206	1.75
H	31	1206	1.9
K	21	0805	1.4
X	05	0402	0.22
Y	03	0201	0.22
	05	0402	0.33
Z	05	0402	0.5

④ Dielectric Code

Temperature-Compensating type			
Code	Temperature Range(°C)	ppm/°C	
CG	-55 to 125	0	±30
CH			±60

• All parts of C0G will be marked as "CG" but will conform to the above table.
• Temperature coefficients are determined by calculation based on measurement at 20°C and 85°C.

High Dielectric Constant Type			
Code	Temperature Range(°C)	ΔC (%)	Reference Temp.°C
R5	-55 to 85	±15	25
S6	-55 to 105	±22	
T6		+22/-33	
R7		±15	
K7*	-55 to 125	±15	
S7		±22	
T7		+22/-33	

*Special spec: Change in capacitance under 50% of rated voltage applied.

⑤ Voltage Code

Code	Rated Voltage
0E	2.5 Vdc
0G	4 Vdc
0J	6.3 Vdc
1A	10 Vdc
1C	16 Vdc
1E	25 Vdc
1V	35 Vdc
1H	50 Vdc
2A	100 Vdc

*Products of identical size and capacitance may sometimes be substituted with a higher rated voltage type.

⑦ Tolerance Code

Code	Tolerance
A	±0.05 pF
B	±0.1 pF
C	±0.25 pF
D	±0.5 pF
G	±2 %
J	±5 %
K	±10 %
M	±20 %

⑥ Capacitance Code (Example)

Code	Capacitance
R50	0.5 pF
1R0	1 pF
100	10 pF
101	100 pF
102	1 nF
103	10 nF
104	0.1 μF
105	1 μF
106	10 μF
107	100 μF

Capacitance expressed in pF.
Two significant digits plus number of zeros.
Example:
102 = 10 (significant digits) followed by 2 zeros
= 1,000 pF = 1 nF.

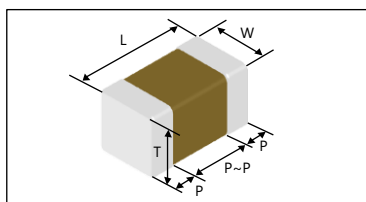
For Values < 10pF,
Letter R denotes decimal point,
Example:
1R0 = 1.0 pF.

⑧ Packaging Code

Code	Size Code	Pitch	Width	Material	Reel size
T	15 to 31	4mm	8mm	Paper	φ180
H	02 to 05	2mm			
Q	03	1mm			
U	21 to 32	4mm	8mm	Plastic	φ330
P	02	1mm	4mm		
M	15 / 21	4mm	8mm	Paper	
N	02 to 05	2mm			
W	03	1mm			
L	21 to 32	4mm	8mm	Plastic	

Multilayer Ceramic Chip Capacitors Dimension

■ Two Terminal Capacitors (KGM/KGT/KGU/KAM Series) ■ Packaging Code



20kp (E 8 / 2)

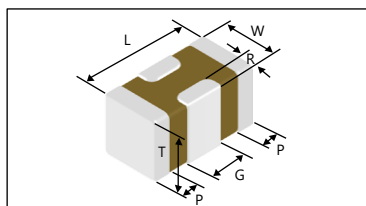
kp means 1000pieces

Taping Material		Taping Width		Pitch	
Code	Material	Code	Width	Code	Pitch
E	Plastic	4	4mm	1	1mm
P	Paper	8	8mm	2	2mm
				4	4mm

Size	Code		Thickness Code	Dimension (mm)						Quantity per reel	
	EIA	JIS		L	W	T	P min.	P max.	P ~ P min.	φ180 Reel	φ330 Reel
02	01005	0402	A	0.4 ±0.02	0.2 ±0.02	0.2 ±0.02	0.07	0.14	0.13	40kp (E4/1)	-
										20kp (P8/2)	80kp (P8/2)
03	0201	0603	Y	0.6 ±0.03	0.3 ±0.03	0.22 max.	0.1	0.2	0.2	30kp (P8/1)	150kp (P8/1)
			A			0.3 ±0.03				15kp (P8/2)	50kp (P8/2)
			B	0.6 ±0.05	0.3 ±0.05	0.3 ±0.05	0.13	0.23	0.19	15kp (P8/2)	50kp (P8/2)
			Y*	0.6 ±0.09	0.3 ±0.09	0.22 max.	0.13	0.23	0.19	15kp (P8/2)	50kp (P8/2)
			C			0.3 ±0.09				10kp (P8/2)	-
			D			0.6 ±0.09				0.3 ±0.09	0.5 ±0.05
			D*	0.1	0.25		0.19				
05	0402	1005	Y	1.0 ±0.05	0.5 ±0.05	0.33 max.	0.15	0.35	0.3	10kp (P8/2)	50kp (P8/2)
			A			0.5 ±0.05				10kp (P8/2)	50kp (P8/2)
			X	1.0 ±0.1	0.5 ±0.05	0.22 max.	0.15	0.35	0.3	10kp (P8/2)	50kp (P8/2)
			B	1.0 ±0.15	0.5 ±0.15	0.5 ±0.15				10kp (P8/2)	40kp (P8/2)
			Y*	1.0 ±0.2	0.5 ±0.2	0.33 max.	0.15	0.35	0.3	10kp (P8/2)	-
			Z			0.5 max.				50kp (P8/2)	
			A*			0.55 max.				50kp (P8/2)	
			C	1.0 ±0.2	0.5 ±0.2	0.5 ±0.2	0.15	0.35	0.3	10kp (P8/2)	40kp (P8/2)
			D			0.8 max.				30kp (P8/2)	
15	0603	1608	A	1.6 ±0.1	0.8 ±0.1	0.8 ±0.1	0.2	0.6	0.5	4kp (P8/4)	10kp (P8/4)
			C	1.6 ±0.2	0.8 ±0.2	0.8 ±0.2					
21	0805	2012	K	2.0 ±0.15	1.25 ±0.15	1.25 ±0.15	0.2	0.75	0.7	3kp (E8/4)	10kp (E8/4)
			A	2.0 ±0.2	1.25 ±0.2	1.25 ±0.2					
31	1206	3216	F	3.2 ±0.2	1.6 ±0.15	1.6 ±0.15	0.3	0.85	1.4	2.5kp (E8/4)	5kp (E8/4)
			A		1.6 ±0.2	1.6 ±0.2					
			H	3.2 ±0.3	1.6 ±0.3	1.6 ±0.3	0.3	0.85	1.9	2kp (E8/4)	-
32	1210	3225	A	3.2 ±0.3	2.5 ±0.2	2.5 ±0.2	0.3	1.0	1.4	1kp (E8/4)	4kp (E8/4)

If there is a "" in the thickness code indicates the same thickness (T-dimension) but different L/W or P-dimension.

■ Three Terminal Capacitors (KGN Series)



Size	Code		Thickness Code	Dimension (mm)						Quantity per reel	
	EIA	JIS		L	W	T	G	P	R	φ180 Reel	φ330 Reel
05	0402	1005	Z	1.0 ±0.1	0.5 ±0.2	0.5 max.	0.3±0.1	0.15±0.1	≥0.05	10kp (P8/2)	-
			B	1.0 ±0.15	0.5 ±0.15	0.5 ±0.15					
			C	1.0 ±0.2	0.5 ±0.2	0.5 ±0.2					

■ tanδ Code

tanδ Code	tanδ max.
3	5 %
5	7.5 %
7	10 %
8	12.5 %
9	15 %
10	20 %

Multilayer Ceramic Chip Capacitors

General

KGM Series



■ Features

We offer a diverse product line ranging from ultra-compact (0.4×0.2mm) to large (3.2×2.5mm) components configured for a variety of temperature characteristics, rated voltages, and packages. We offer the choice and flexibility for almost any applications.

■ Applications

This standard type is ideal for use in a wide range of applications, from commercial to industrial equipment.

Class1 : Temperature-Compensating type
CG / CH (JIS)

● Capacitance chart ■ Standard Spec.1

EIA Size (Series Code)	Voltage [Vdc] (Code)	CG/CH (JIS)																										
		Capacitance (Code)																										
		1pF (1R0)	1.5pF (1R5)	2pF (2R0)	3pF (3R0)	4pF (4R0)	5pF (5R0)	6pF (6R0)	7pF (7R0)	8pF (8R0)	9pF (9R0)	10pF (100)	12pF (120)	15pF (150)	18pF (180)	22pF (220)	27pF (270)	33pF (330)	39pF (390)	47pF (470)	56pF (560)	68pF (680)	82pF (820)	100pF (101)	120pF (121)	150pF (151)	180pF (181)	220pF (221)
01005 (KGM02)	50 (1H)					A												A									A	
	25 (1E)					A												A									A	
	16 (1C)					A												A									A	

Class2 : High Dielectric Constant type
X5R (EIA) [R5]

● Capacitance chart ■ Standard Spec.1 ■ Standard Spec.2

EIA Size (Series Code)	Voltage [Vdc] (Code)	X5R (EIA) [R5]																				
		Capacitance (Code)																				
		100pF (101)	150pF (151)	220pF (221)	330pF (331)	470pF (471)	680pF (681)	1nF (102)	1.5nF (152)	2.2nF (222)	3.3nF (332)	4.7nF (472)	6.8nF (682)	10nF (103)	15nF (153)	22nF (223)	33nF (333)	47nF (473)	68nF (683)	0.1μF (104)	0.22μF (224)	0.47μF (474)
01005 (KGM02)	16 (1C)				A8						A8			A8								
	10 (1A)				A8						A8			A8						A8		
	6.3 (0J)																A8			A8		A8

EIA Size (Series Code)	Voltage [Vdc] (Code)	X5R (EIA) [R5]																	
		Capacitance (Code)																	
		22nF (223)	33nF (333)	47nF (473)	68nF (683)	0.1μF (104)	0.22μF (224)	0.47μF (474)	1μF (105)	2.2μF (225)	4.7μF (475)	10μF (106)	15μF (156)	22μF (226)	47μF (476)	100μF (107)			
0201 (KGM03)	25 (1E)						C8												
	16 (1C)						C8		C10										
	10 (1A)	A7	A7	A7	A7	A7	A7	B8	C10	C9									
	6.3 (0J)						A8	B8	B8	C8									
	4 (0G)										C8	D*7							
0402 (KGM05)	35 (1V)								A7	C8									
	25 (1E)								A7	A*8	C8								
	16 (1C)						A8	A7	A8	C8									
	10 (1A)						A8	A7	A8	B8	C8								
	6.3 (0J)								A8			C8	B8	C8	D8				
0603 (KGM15)	35 (1V)										C8	C9							
	25 (1E)										C8	C9			C8				
	16 (1C)											C9			C8				
	10 (1A)														C8				
	6.3 (0J)															C8			
0805 (KGM21)	35 (1V)																		
	25 (1E)														A8				
	16 (1C)														A8				
	10 (1A)														A8				
	6.3 (0J)															A7	A8		
1206 (KGM31)	35 (1V)																		
	25 (1E)																		
	16 (1C)																		
	10 (1A)																		
	6.3 (0J)																		
1210 (KGM32)	35 (1V)																		
	25 (1E)																		
	16 (1C)																		
	10 (1A)																		
	6.3 (0J)																		

If there is a "" in the thickness code indicates the same thickness (T-dimension) but different L/W or P-dimension.

The code in the Capacitance chart represents product thickness (T-dimension) and tan delta.
For details, please refer to the "Part number list".

Temperature-Compensating type : [KGM series Class1 : Temperature-Compensating type CG / CH(JIS)]
High Dielectric Constant type : [KGM series Class2 : High Dielectric Constant type X5R(EIA) [R5]]

(Example) In case of "A" for KGM02;T : 0.2±0.02mm

In case of "A7" for KGM03;T : 0.3±0.03mm, tanδ: 10.0% max.

For test items, specifications, and measurement conditions, please refer to "Test Conditions and Standards".

Temperature-Compensating type : Standard Spec.1 : [Temperature-Compensating type(Class1) KGM/KGU Series (Standard Spec.1)]
High Dielectric Constant type : Standard Spec.1 : [High Dielectric Constant type(Class2) KGM/KGT Series (Standard Spec.1)]
Standard Spec.2 : [High Dielectric Constant type(Class2) KGM/KGT Series (Standard Spec.2)]

Please contact for capacitance value other than standard.

Class2 : High Dielectric Constant type
X6S (EIA) [S6]

● Capacitance chart ■ Standard Spec .2

X6S (EIA) [S6]											
EIA Size (Series Code)	Voltage [Vdc] (Code)	Capacitance (Code)									
		0.1μF (104)	0.22μF (224)	0.47μF (474)	1μF (105)	2.2μF (225)	4.7μF (475)	10μF (106)	22μF (226)	47μF (476)	100μF (107)
0201 (KGM03)	10 (1A)				C10						
	6.3 (0J)				C10						
	4 (0G)				C10						
	2.5 (0E)						D9				
0402 (KGM05)	25 (1E)				A8						
	16 (1C)			A8	A*8						
	10 (1A)			A8	A8		C8				
	6.3 (0J)					B8	C8	C8			
0603 (KGM15)	4 (0G)								D8		
	16 (1C)						C8	C9			
	10 (1A)							C9	C8		
	6.3 (0J)								C8		
0805 (KGM21)	4 (0G)								C8	C8	
	2.5 (0E)									C8	C8
	25 (1E)						K8	A8			
	16 (1C)							A8			
	10 (1A)							A8			
	6.3 (0J)							A8			
	4 (0G)								A7	A8	

If there is a "" in the thickness code indicates the same thickness (T-dimension) but different L/W or P-dimension.

Class2 : High Dielectric Constant type
X6T (EIA) [T6]

● Capacitance chart ■ Standard Spec .2

X6T (EIA) [T6]							
EIA Size (Series Code)	Voltage [Vdc] (Code)	Capacitance (Code)					
		0.1μF (104)	0.22μF (224)	0.47μF (474)	1μF (105)	2.2μF (225)	4.7μF (475)
0201 (KGM03)	10 (1A)		C8				
	4 (0G)				B8	C8	
	2.5 (0E)						

Class2 : High Dielectric Constant type
X7R (EIA) [R7]

● Capacitance chart ■ Standard Spec.1 ■ Standard Spec .2

X7R (EIA) [R7]										
EIA Size (Series Code)	Voltage [Vdc] (Code)	Capacitance (Code)								
		100pF (101)	150pF (151)	220pF (221)	330pF (331)	470pF (471)	680pF (681)	1nF (102)	1.5nF (152)	2.2nF (222)
01005 (KGM02)	16 (1C)	A8						A8		

X7R (EIA) [R7]									
EIA Size (Series Code)	Voltage [Vdc] (Code)	Capacitance (Code)							
		0.1μF (104)	0.22μF (224)	0.47μF (474)	1μF (105)	2.2μF (225)	4.7μF (475)	10μF (106)	22μF (226)
0402 (KGM05)	25 (1E)	A8							
	6.3 (0J)			A8	A8				
0603 (KGM15)	25 (1E)				A7				
	10 (1A)					A7			
0805 (KGM21)	50 (1H)				A3				
	25 (1E)						A7		
	16 (1C)						K7	A8	
	6.3 (0J)							A8	
	50 (1H)						A3		
	25 (1E)						A3	A3	
1206 (KGM31)	16 (1C)							A8	
	10 (1A)								A5
	6.3 (0J)								A8
1210 (KGM32)	50 (1H)							A3	
	25 (1E)							A8	

The code in the Capacitance chart represents product thickness (T-dimension) and tan delta.
For details, please refer to the "Part number list".

High Dielectric Constant type : S6 Dielectric : [KGM series Class2 : High Dielectric Constant type X6S(EIA) [S6]]
T6 Dielectric : [KGM series Class2 : High Dielectric Constant type X5R(EIA) [T6]]
R7 Dielectric : [KGM series Class2 : High Dielectric Constant type X5R(EIA) [R7]]

(Example) In case of "C9" for KGM15;T: 0.8±0.2mm, tanδ: 15.0% max.

For test items, specifications, and measurement conditions, please refer to "Test Conditions and Standards".

High Dielectric Constant type : Standard Spec.1 : [High Dielectric Constant type(Class2) KGM/KGT Series (Standard Spec.1)]
Standard Spec.2 : [High Dielectric Constant type(Class2) KGM/KGT Series (Standard Spec.2)]

Please contact for capacitance value other than standard.

Multilayer Ceramic Chip Capacitors

General

KGM Series



Class2 : High Dielectric Constant type
X7R (KAVX) [K7]

● Capacitance chart ■ Standard Spec .2

X7R (KAVX) [K7]									
EIA Size (Series Code)	Voltage [Vdc] (Code)	Capacitance (Code)							
		0.1μF (104)	0.22μF (224)	0.47μF (474)	1μF (105)	2.2μF (225)	4.7μF (475)	10μF (106)	22μF (226)
0402 (KGM05)	10 (1A)					B3			
0603 (KGM15)	16 (1C)						C8		
	10 (1A)							C9	
0805 (KGM21)	10 (1A)								A8
1206 (KGM31)	100 (2A)					A3	H3		
	50 (1H)					A3			

Special spec : Change in capacitance under 50% of rated voltage applied.
Please Refer to "How to Order", "④ Dielectric Code" for details.

Class2 : High Dielectric Constant type
X7S (EIA) [S7]

● Capacitance chart ■ Standard Spec.1 ■ Standard Spec .2

X7S (EIA) [S7]									
EIA Size (Series Code)	Voltage [Vdc] (Code)	Capacitance (Code)							
		0.1μF (104)	0.22μF (224)	0.47μF (474)	1μF (105)	2.2μF (225)	4.7μF (475)		
0201 (KGM03)	6.3 (0J)	A7							
0402 (KGM05)	10 (1A)					B3			
	6.3 (0J)				A8	B3			
	4 (0G)				A8	B3			
0805 (KGM21)	100 (2A)				A3				
1206 (KGM31)	100 (2A)					A3	H3		

Class2 : High Dielectric Constant type
X7T (EIA) [T7]

● Capacitance chart ■ Standard Spec .2

X7T (EIA) [T7]									
EIA Size (Series Code)	Voltage [Vdc] (Code)	Capacitance (Code)							
		0.1μF (104)	0.22μF (224)	0.47μF (474)	1μF (105)	2.2μF (225)	4.7μF (475)	10μF (106)	22μF (226)
0201 (KGM03)	6.3 (0J)				C8				
0402 (KGM05)	10 (1A)						C8		
0603 (KGM15)	16 (1C)						C8		
	10 (1A)						C8	C9	
	6.3 (0J)						C8	C9	
0805 (KGM21)	10 (1A)								A8
	6.3 (0J)								A8

The code in the Capacitance chart represents product thickness (T-dimension) and tan delta.
For details, please refer to the "Part number list".

High Dielectric Constant type : K7 Dielectric : [KGM series Class2 : High Dielectric Constant type X7R(KAVX) [K7]]
S7 Dielectric : [KGM series Class2 : High Dielectric Constant type X7S(EIA) [S7]]
T7 Dielectric : [KGM series Class2 : High Dielectric Constant type X7T(EIA) [T7]]

(Example) In case of "C9" for KGM15;T: 0.8±0.2mm, tanδ: 15.0% max.

For test items, specifications, and measurement conditions, please refer to "Test Conditions and Standards".

High Dielectric Constant type : Standard Spec.1 : [High Dielectric Constant type(Class2) KGM/KGT Series (Standard Spec.1)]
Standard Spec.2 : [High Dielectric Constant type(Class2) KGM/KGT Series (Standard Spec.2)]

Please contact for capacitance value other than standard.

Multilayer Ceramic Chip Capacitors



General

KGM Series



Class1 : Temperature-Compensating type
CG / CH (JIS)

Part number list General KGM02 Series Temperature Characteristic: CA: CG/CH Tolerance □: B: ± 0.1 pF/ C: ± 0.25 pF/ D: ± 0.5 pF/ J: $\pm 5\%$ / K: $\pm 10\%$
■ Standard Spec.1

EIA Size (Series Code)	Voltage [Vdc]	Capacitance	Tolerance Code :□	Part Number	Thickness Code	Dimension[mm]						Q min.	Load Life	Packaging Code (QTY):#	
						L	W	T	P min.	P max.	P ~ P min.			φ180 Reel	φ330 Reel
01005 (KGM02)	50	1 pF	B/C	KGM02ACΔ1H1R0□#	A	0.4±0.02	0.2±0.02	0.2±0.02	0.07	0.14	0.13	400+20C	×2.0	H(20kp) / P(40kp)	N(80kp)
		1.5 pF	B/C	KGM02ACΔ1H1R5□#											
		2 pF	B/C	KGM02ACΔ1H2R0□#											
		3 pF	B/C	KGM02ACΔ1H3R0□#											
		4 pF	B/C	KGM02ACΔ1H4R0□#											
		5 pF	B/C	KGM02ACΔ1H5R0□#	A	0.4±0.02	0.2±0.02	0.2±0.02	0.07	0.14	0.13	400+20C	×2.0	H(20kp) / P(40kp)	N(80kp)
		6 pF	C/D	KGM02ACΔ1H6R0□#											
		7 pF	C/D	KGM02ACΔ1H7R0□#											
		8 pF	C/D	KGM02ACΔ1H8R0□#											
		9 pF	C/D	KGM02ACΔ1H9R0□#											
		10 pF	J/K	KGM02ACΔ1H100□#	A	0.4±0.02	0.2±0.02	0.2±0.02	0.07	0.14	0.13	400+20C	×2.0	H(20kp) / P(40kp)	N(80kp)
		12 pF	J/K	KGM02ACΔ1H120□#											
		15 pF	J/K	KGM02ACΔ1H150□#											
		18 pF	J/K	KGM02ACΔ1H180□#											
		22 pF	J/K	KGM02ACΔ1H220□#											
		27 pF	J/K	KGM02ACΔ1H270□#	A	0.4±0.02	0.2±0.02	0.2±0.02	0.07	0.14	0.13	1000	×2.0	H(20kp) / P(40kp)	N(80kp)
		33 pF	J/K	KGM02ACΔ1H330□#											
		39 pF	J/K	KGM02ACΔ1H390□#											
		47 pF	J/K	KGM02ACΔ1H470□#											
		56 pF	J/K	KGM02ACΔ1H560□#											
		68 pF	J/K	KGM02ACΔ1H680□#	A	0.4±0.02	0.2±0.02	0.2±0.02	0.07	0.14	0.13	1000	×2.0	H(20kp) / P(40kp)	N(80kp)
		82 pF	J/K	KGM02ACΔ1H820□#											
		100 pF	J/K	KGM02ACΔ1H101□#											
		120 pF	J/K	KGM02ACΔ1H121□#											
		150 pF	J/K	KGM02ACΔ1H151□#											
		180 pF	J/K	KGM02ACΔ1H181□#											
		220 pF	J/K	KGM02ACΔ1H221□#											

For test items, specifications, and measurement conditions, please refer to "Test Conditions and Standards".
Temperature-Compensating type : Standard Spec.1 : [Temperature-Compensating type(Class1) KGM/KGU Series (Standard Spec.1)]

Load Life: Applied voltage in the load life test, expressed as a multiple of the rated voltage.

Q: "C" in "400 + 20C min" denotes the nominal capacitance value (pF).

(Example): For "10pF", $400 + 20 \times 10 = 600$.

Please contact for capacitance value other than standard.

Multilayer Ceramic Chip Capacitors



General

KGM Series



Class1 : Temperature-Compensating type
CG / CH (JIS)

Part number list

General KGM02 Series Temperature Characteristic: CA: CG/CH Tolerance □: B: ±0.1 pF/ C: ±0.25 pF/ D: ± 0.5 pF/ J: ± 5%/ K: ±10%

■ Standard Spec.1

EIA Size (Series Code)	Voltage [Vdc]	Capacitance	Tolerance Code :□	Part Number	Thickness Code	Dimension[mm]						Q min.	Load Life	Packaging Code (QTY):#	
						L	W	T	P min.	P max.	P ~ P min.			φ180 Reel	φ330 Reel
01005 (KGM02)	25	1 pF	B/C	KGM02ACΔ1E1R0□#	A	0.4±0.02	0.2±0.02	0.2±0.02	0.07	0.14	0.13	400+20C	×2.0	H(20kp) / P(40kp)	N(80kp)
		1.5 pF	B/C	KGM02ACΔ1E1R5□#											
		2 pF	B/C	KGM02ACΔ1E2R0□#											
		3 pF	B/C	KGM02ACΔ1E3R0□#											
		4 pF	B/C	KGM02ACΔ1E4R0□#											
		5 pF	B/C	KGM02ACΔ1E5R0□#											
		6 pF	C/D	KGM02ACΔ1E6R0□#	A	0.4±0.02	0.2±0.02	0.2±0.02	0.07	0.14	0.13	400+20C	×2.0	H(20kp) / P(40kp)	N(80kp)
		7 pF	C/D	KGM02ACΔ1E7R0□#											
		8 pF	C/D	KGM02ACΔ1E8R0□#											
		9 pF	C/D	KGM02ACΔ1E9R0□#											
		10 pF	J/K	KGM02ACΔ1E100□#	A	0.4±0.02	0.2±0.02	0.2±0.02	0.07	0.14	0.13	400+20C	×2.0	H(20kp) / P(40kp)	N(80kp)
		12 pF	J/K	KGM02ACΔ1E120□#											
		15 pF	J/K	KGM02ACΔ1E150□#											
		18 pF	J/K	KGM02ACΔ1E180□#											
		22 pF	J/K	KGM02ACΔ1E220□#											
		27 pF	J/K	KGM02ACΔ1E270□#											
		33 pF	J/K	KGM02ACΔ1E330□#	A	0.4±0.02	0.2±0.02	0.2±0.02	0.07	0.14	0.13	1000	×2.0	H(20kp) / P(40kp)	N(80kp)
		39 pF	J/K	KGM02ACΔ1E390□#											
		47 pF	J/K	KGM02ACΔ1E470□#											
		56 pF	J/K	KGM02ACΔ1E560□#											
		68 pF	J/K	KGM02ACΔ1E680□#											
		82 pF	J/K	KGM02ACΔ1E820□#											
		100 pF	J/K	KGM02ACΔ1E101□#	A	0.4±0.02	0.2±0.02	0.2±0.02	0.07	0.14	0.13	1000	×2.0	H(20kp) / P(40kp)	N(80kp)
		120 pF	J/K	KGM02ACΔ1E121□#											
		150 pF	J/K	KGM02ACΔ1E151□#											
		180 pF	J/K	KGM02ACΔ1E181□#											
		220 pF	J/K	KGM02ACΔ1E221□#											
	16	1 pF	B/C	KGM02ACΔ1C1R0□#	A	0.4±0.02	0.2±0.02	0.2±0.02	0.07	0.14	0.13	400+20C	×2.0	H(20kp) / P(40kp)	N(80kp)
		1.5 pF	B/C	KGM02ACΔ1C1R5□#											
		2 pF	B/C	KGM02ACΔ1C2R0□#											
		3 pF	B/C	KGM02ACΔ1C3R0□#											
		4 pF	B/C	KGM02ACΔ1C4R0□#											
		5 pF	B/C	KGM02ACΔ1C5R0□#											
		6 pF	C/D	KGM02ACΔ1C6R0□#	A	0.4±0.02	0.2±0.02	0.2±0.02	0.07	0.14	0.13	400+20C	×2.0	H(20kp) / P(40kp)	N(80kp)
		7 pF	C/D	KGM02ACΔ1C7R0□#											
		8 pF	C/D	KGM02ACΔ1C8R0□#											
		9 pF	C/D	KGM02ACΔ1C9R0□#											
		10 pF	J/K	KGM02ACΔ1C100□#	A	0.4±0.02	0.2±0.02	0.2±0.02	0.07	0.14	0.13	400+20C	×2.0	H(20kp) / P(40kp)	N(80kp)
		12 pF	J/K	KGM02ACΔ1C120□#											
		15 pF	J/K	KGM02ACΔ1C150□#											
		18 pF	J/K	KGM02ACΔ1C180□#											
		22 pF	J/K	KGM02ACΔ1C220□#											
		27 pF	J/K	KGM02ACΔ1C270□#											
		33 pF	J/K	KGM02ACΔ1C330□#	A	0.4±0.02	0.2±0.02	0.2±0.02	0.07	0.14	0.13	1000	×2.0	H(20kp) / P(40kp)	N(80kp)
		39 pF	J/K	KGM02ACΔ1C390□#											
		47 pF	J/K	KGM02ACΔ1C470□#											
		56 pF	J/K	KGM02ACΔ1C560□#											
		68 pF	J/K	KGM02ACΔ1C680□#											
		82 pF	J/K	KGM02ACΔ1C820□#											
		100 pF	J/K	KGM02ACΔ1C101□#	A	0.4±0.02	0.2±0.02	0.2±0.02	0.07	0.14	0.13	1000	×2.0	H(20kp) / P(40kp)	N(80kp)
		120 pF	J/K	KGM02ACΔ1C121□#											
		150 pF	J/K	KGM02ACΔ1C151□#											
		180 pF	J/K	KGM02ACΔ1C181□#											
		220 pF	J/K	KGM02ACΔ1C221□#											

For test items, specifications, and measurement conditions, please refer to "Test Conditions and Standards".

Temperature-Compensating type : Standard Spec.1 : [Temperature-Compensating type(Class1) KGM/KGU Series (Standard Spec.1)]

Load Life: Applied voltage in the load life test, expressed as a multiple of the rated voltage.

Q: "C" in "400 + 20C min" denotes the nominal capacitance value (pF).

(Example) : For "10pF", 400 + 20 × 10 = 600.

Please contact for capacitance value other than standard.

Multilayer Ceramic Chip Capacitors

General

KGM Series



Class2 : High Dielectric Constant type
X5R (EIA) [R5]

Part number list General KGM02 to 03 Series Temperature Characteristic: R5 Tolerance □: K: ±10%/ M: ±20%
■ Standard Spec.1 ■ Standard Spec.2

EIA Size (Series Code)	Voltage [Vdc]	Capacitance	Tolerance Code :□	Part Number	Thickness /tanδ Code	Dimension[mm]						tanδ max. [%]	Load Life	Packaging Code (QTY):#			
						L	W	T	P min.	P max.	P ~ P min.			φ180 Reel	φ330 Reel		
01005 (KGM02)	16	100 pF	K/M	KGM02AR51C101□#	A8	0.4±0.02	0.2±0.02	0.2±0.02	0.07	0.14	0.13	12.5	×1.5	H(20kp) / P(40kp)	N(80kp)		
		150 pF	K/M	KGM02AR51C151□#													
		220 pF	K/M	KGM02AR51C221□#													
		330 pF	K/M	KGM02AR51C331□#													
		470 pF	K/M	KGM02AR51C471□#													
		680 pF	K/M	KGM02AR51C681□#													
		1 nF	K/M	KGM02AR51C102□#	A8	0.4±0.02	0.2±0.02	0.2±0.02	0.07	0.14	0.13	12.5	×1.5	H(20kp) / P(40kp)	N(80kp)		
		1.5 nF	K/M	KGM02AR51C152□#													
		2.2 nF	K/M	KGM02AR51C222□#													
		3.3 nF	K/M	KGM02AR51C332□#													
		4.7 nF	K/M	KGM02AR51C472□#													
		6.8 nF	K/M	KGM02AR51C682□#													
		10 nF	K/M	KGM02AR51C103□#	A8	0.4±0.02	0.2±0.02	0.2±0.02	0.07	0.14	0.13	12.5	×1.5	H(20kp) / P(40kp)	N(80kp)		
	150 pF	K/M	KGM02AR51A151□#														
	220 pF	K/M	KGM02AR51A221□#														
	330 pF	K/M	KGM02AR51A331□#														
	470 pF	K/M	KGM02AR51A471□#														
	680 pF	K/M	KGM02AR51A681□#														
	1 nF	K/M	KGM02AR51A102□#	A8	0.4±0.02	0.2±0.02	0.2±0.02	0.07	0.14	0.13	12.5	×1.5	H(20kp) / P(40kp)	N(80kp)			
	1.5 nF	K/M	KGM02AR51A152□#														
	2.2 nF	K/M	KGM02AR51A222□#														
	3.3 nF	K/M	KGM02AR51A332□#														
	4.7 nF	K/M	KGM02AR51A472□#														
	6.8 nF	K/M	KGM02AR51A682□#														
	10 nF	K/M	KGM02AR51A103□#	A8	0.4±0.02	0.2±0.02	0.2±0.02	0.07	0.14	0.13	12.5	×1.5	H(20kp) / P(40kp)	N(80kp)			
	0.1 μF	K/M	KGM02AR51A104□#	A8	0.4±0.02	0.2±0.02	0.2±0.02	0.07	0.14	0.13	12.5	×1.0	H(20kp) / P(40kp)	N(80kp)			
	22 nF	K/M	KGM02AR50J223□#														
	33 nF	K/M	KGM02AR50J333□#														
	47 nF	K/M	KGM02AR50J473□#														
	68 nF	K/M	KGM02AR50J683□#														
	0.1 μF	K/M	KGM02AR50J104□#	A8	0.4±0.02	0.2±0.02	0.2±0.02	0.07	0.14	0.13	12.5	×1.3	H(20kp) / P(40kp)	N(80kp)			
	0.22 μF	K/M	KGM02AR50J224□#	A8	0.4±0.02	0.2±0.02	0.2±0.02	0.07	0.14	0.13	12.5	×1.0	H(20kp) / P(40kp)	N(80kp)			
	0.47 μF	M	KGM02AR50J474M#														
	0201 (KGM03)	25	0.22 μF	K/M	KGM03CR51E224□#	C8	0.6±0.09	0.3±0.09	0.3±0.09	0.13	0.23	0.19	12.5	×1.0	H(15kp)	N(50kp)	
			0.22 μF	K/M	KGM03CR51C224□#	C8	0.6±0.09	0.3±0.09	0.3±0.09	0.13	0.23	0.19	12.5	×1.0	H(15kp)	N(50kp)	
1 μF			K/M	KGM03CR51C105□#	C10	0.6±0.09	0.3±0.09	0.3±0.09	0.13	0.23	0.19	20.0	×1.0	H(15kp)	N(50kp)		
33 nF		K/M	KGM03AR51A333□#														
47 nF		K/M	KGM03AR51A473□#														
68 nF		K/M	KGM03AR51A683□#														
0.1 μF		K/M	KGM03AR51A104□#	A7	0.6±0.03	0.3±0.03	0.3±0.03	0.1	0.2	0.2	10.0	×1.3	H(15kp) / Q(30kp)	N(50kp) / W(150kp)			
0.22 μF		K/M	KGM03AR51A224□#														
1 μF		K/M	KGM03BR51A105□#									B8	0.6±0.05	0.3±0.05	0.3±0.05	0.13	0.23
			KGM03CR51A105□#	C10	0.6±0.09	0.3±0.09	0.3±0.09	0.13	0.23	0.19							
2.2 μF		K/M	KGM03CR51A225□#	C9	0.6±0.09	0.3±0.09	0.3±0.09	0.13	0.23	0.19	15.0						
1 μF		K/M	KGM03BR50J105□#	B8	0.6±0.05	0.3±0.05	0.3±0.05	0.13	0.23	0.19	12.5	×1.2	H(15kp) / Q(30kp)	N(50kp) / W(150kp)			
2.2 μF		M	KGM03BR50J225M#												12.5		
		K/M	KGM03CR50J225□#								C8	0.6±0.09	0.3±0.09	0.3±0.09	0.13	0.23	0.19
4.7 μF		M	KGM03DR50J475MH	D9	0.6±0.09	0.3±0.09	0.5±0.05	0.13	0.23	0.19							
10 μF		M	KGM03DR50G106MH	D*7	0.6±0.09	0.3±0.09	0.5±0.05	0.1	0.25	0.19	10.0	×1.0	H(10kp)	—(—)			

If there is a "" in the thickness code indicates the same thickness (T-dimension) but different L/W or P-dimension.

For test items, specifications, and measurement conditions, please refer to "Test Conditions and Standards".

High Dielectric Constant type : Standard Spec.1 : [High Dielectric Constant type(Class2) KGM/KGT Series (Standard Spec.1)]

Standard Spec.2 : [High Dielectric Constant type(Class2) KGM/KGT Series (Standard Spec.2)]

Load Life: Applied voltage in the load life test, expressed as a multiple of the rated voltage.

Please contact for capacitance value other than standard.

Multilayer Ceramic Chip Capacitors



General

KGM Series



Class2 : High Dielectric Constant type
X5R (EIA) [R5]

Part number list General KGM05 to 32 Series Temperature Characteristic: R5 Tolerance □: K: ±10%/ M: ±20%
 ■ Standard Spec.1 ■ Standard Spec.2

EIA Size (Series Code)	Voltage [Vdc]	Capacitance	Tolerance Code :□	Part Number	Thickness /tanδ Code	Dimension[mm]						tanδ max. [%]	Load Life	Packaging Code (QTY):#	
						L	W	T	P min.	P max.	P ~ P min.			φ180 Reel	φ330 Reel
0402 (KGM05)	35	1 μF	K/M	KGM05AR51V105□#	A7	1.0±0.05	0.5±0.05	0.5±0.05	0.15	0.35	0.3	10.0	×1.0	H(10kp)	N(50kp)
		2.2 μF	M	KGM05CR51V225M#	C8	1.0±0.2	0.5±0.2	0.5±0.2	0.15	0.35	0.3	12.5	×1.0	H(10kp)	N(40kp)
	25	1 μF	K/M	KGM05AR51E105□#	A7	1.0±0.05	0.5±0.05	0.5±0.05	0.15	0.35	0.3	10.0	×1.0	H(10kp)	N(50kp)
		2.2 μF	K/M	KGM05AR51E225□#	A*8	1.0±0.2	0.5±0.2	0.55max.	0.15	0.35	0.3	12.5	×1.0	H(10kp)	N(50kp)
			M	KGM05CR51E225M#	C8	1.0±0.2	0.5±0.2	0.5±0.2	0.15	0.35	0.3	12.5	×1.0	H(10kp)	N(40kp)
	16	4.7 μF	K/M	KGM05CR51E475□#	C8	1.0±0.2	0.5±0.2	0.5±0.2	0.15	0.35	0.3	12.5	×1.0	H(10kp)	N(40kp)
		0.47 μF	K/M	KGM05AR51C474□#	A8	1.0±0.05	0.5±0.05	0.5±0.05	0.15	0.35	0.3	12.5	×1.0	H(10kp)	N(50kp)
		1 μF	K/M	KGM05AR51C105□#	A7	1.0±0.05	0.5±0.05	0.5±0.05	0.15	0.35	0.3	10.0	×1.5	H(10kp)	N(50kp)
		2.2 μF	K/M	KGM05AR51C225□#	A8	1.0±0.05	0.5±0.05	0.5±0.05	0.15	0.35	0.3	12.5	×1.0	H(10kp)	N(50kp)
	10	4.7 μF	K/M	KGM05CR51C475□#	C8	1.0±0.2	0.5±0.2	0.5±0.2	0.15	0.35	0.3	12.5	×1.0	H(10kp)	N(40kp)
		0.47 μF	K/M	KGM05AR51A474□#	A8	1.0±0.05	0.5±0.05	0.5±0.05	0.15	0.35	0.3	12.5	×1.5	H(10kp)	N(50kp)
		1 μF	K/M	KGM05AR51A105□#	A7	1.0±0.05	0.5±0.05	0.5±0.05	0.15	0.35	0.3	10.0	×1.5	H(10kp)	N(50kp)
		2.2 μF	K/M	KGM05AR51A225□#	A8	1.0±0.05	0.5±0.05	0.5±0.05	0.15	0.35	0.3	12.5			
		4.7 μF	K/M	KGM05BR51A475□#	B8	1.0±0.15	0.5±0.15	0.5±0.15	0.15	0.35	0.3	12.5	×1.0	H(10kp)	N(40kp)
				KGM05CR51A475□#	C8	1.0±0.2	0.5±0.2	0.5±0.2	0.15	0.35	0.3	12.5	×1.5		
		10 μF	K/M	KGM05CR51A106□#	C8	1.0±0.2	0.5±0.2	0.5±0.2	0.15	0.35	0.3	12.5	×1.0	H(10kp)	N(40kp)
	6.3	2.2 μF	K/M	KGM05AR50J225□#	A8	1.0±0.05	0.5±0.05	0.5±0.05	0.15	0.35	0.3	12.5	×1.5	H(10kp)	N(50kp)
		10 μF	M	KGM05CR50J106M#	C8	1.0±0.2	0.5±0.2	0.5±0.2	0.15	0.35	0.3	12.5	×1.0	H(10kp)	N(40kp)
		15 μF	M	KGM05BR50J156M#	B8	1.0±0.15	0.5±0.15	0.5±0.15							
		22 μF	M	KGM05CR50J226M#	C8	1.0±0.2	0.5±0.2	0.5±0.2	0.15	0.35	0.3	12.5	×1.0	H(10kp)	N(40kp)
	4			KGM05DR50J226M#	D8	1.0±0.2	0.5±0.2	0.8max.	0.15	0.35	0.3	12.5	×1.0	H(10kp)	N(30kp)
		15 μF	M	KGM05BR50G156M#	B8	1.0±0.15	0.5±0.15	0.5±0.15	0.15	0.35	0.3	12.5	×1.0	H(10kp)	N(40kp)
		22 μF	M	KGM05CR50G226M#	C8	1.0±0.2	0.5±0.2	0.5±0.2							
0603 (KGM15)	35	4.7 μF	K/M	KGM15CR51V475□#	C8	1.6±0.2	0.8±0.2	0.8±0.2	0.2	0.6	0.5	12.5	×1.0	T(4kp)	M(10kp)
		10 μF	M	KGM15CR51V106M#	C9	1.6±0.2	0.8±0.2	0.8±0.2	0.2	0.6	0.5	15.0			
	25	4.7 μF	K/M	KGM15CR51E475□#	C8	1.6±0.2	0.8±0.2	0.8±0.2	0.2	0.6	0.5	12.5	×1.0	T(4kp)	M(10kp)
		10 μF	K/M	KGM15CR51E106□#	C9	1.6±0.2	0.8±0.2	0.8±0.2	0.2	0.6	0.5	15.0	×1.0	T(4kp)	M(10kp)
		22 μF	M	KGM15CR51E226M#	C8	1.6±0.2	0.8±0.2	0.8±0.2	0.2	0.6	0.5	12.5			
	16	10 μF	K/M	KGM15CR51C106□#	C9	1.6±0.2	0.8±0.2	0.8±0.2	0.2	0.6	0.5	15.0	×1.5	T(4kp)	M(10kp)
		22 μF	M	KGM15CR51C226M#	C8	1.6±0.2	0.8±0.2	0.8±0.2	0.2	0.6	0.5	12.5	×1.0		
	10	22 μF	M	KGM15CR51A226M#	C8	1.6±0.2	0.8±0.2	0.8±0.2	0.2	0.6	0.5	12.5	×1.0	T(4kp)	M(10kp)
	6.3	47 μF	M	KGM15CR50J476M#	C8										
	4	47 μF	M	KGM15CR50G476M#	C8										
0805 (KGM21)	25	22 μF	M	KGM21AR51E226M#	A8	2.0±0.2	1.25±0.2	1.25±0.2	0.2	0.75	0.7	12.5	×1.0	U(3kp)	L(10kp)
	16	10 μF	K/M	KGM21AR51C106□#	A8	2.0±0.2	1.25±0.2	1.25±0.2	0.2	0.75	0.7	12.5	×1.5	U(3kp)	L(10kp)
		22 μF	M	KGM21AR51C226M#	A8	2.0±0.2	1.25±0.2	1.25±0.2	0.2	0.75	0.7	12.5	×1.0		
	10	22 μF	M	KGM21AR51A226M#	A8	2.0±0.2	1.25±0.2	1.25±0.2	0.2	0.75	0.7	12.5	×1.5	U(3kp)	L(10kp)
	6.3	47 μF	M	KGM21AR50J476M#	A7	2.0±0.2	1.25±0.2	1.25±0.2	0.2	0.75	0.7	10.0	×1.0	U(3kp)	L(10kp)
		100 μF	M	KGM21AR50J107M#	A7	2.0±0.2	1.25±0.2	1.25±0.2	0.2	0.75	0.7	12.5			
	4	100 μF	M	KGM21AR50G107M#	A8	2.0±0.2	1.25±0.2	1.25±0.2	0.2	0.75	0.7	12.5	×1.0	U(3kp)	L(10kp)
1206 (KGM31)	100	2.2 μF	K/M	KGM31AR52A225□#	A3	3.2±0.2	1.6±0.2	1.6±0.2	0.3	0.85	1.4	5.0	×1.0	U(2.5kp)	L(5kp)
	50	4.7 μF	K/M	KGM31AR51H475□#	A3	3.2±0.2	1.6±0.2	1.6±0.2	0.3	0.85	1.4	5.0	×1.5		
	25	2.2 μF	K/M	KGM31FR51E225□#	F3	3.2±0.2	1.6±0.15	1.6±0.15	0.3	0.85	1.4	5.0	×1.5	U(2.5kp)	L(5kp)
		10 μF	K/M	KGM31AR51E106□#	A8	3.2±0.2	1.6±0.2	1.6±0.2	0.3	0.85	1.4	12.5			
	16	22 μF	K/M	KGM31AR51C226□#	A8	3.2±0.2	1.6±0.2	1.6±0.2	0.3	0.85	1.4	12.5	×1.5	U(2.5kp)	L(5kp)
1210 (KGM32)	50	10 μF	K/M	KGM32AR51H106□#	A3	3.2±0.3	2.5±0.2	2.5±0.2	0.3	1.0	1.4	5.0	×1.5	U(1kp)	L(4kp)
	35	10 μF	K/M	KGM32AR51V106□#											
	25	10 μF	K/M	KGM32AR51E106□#	A8	3.2±0.3	2.5±0.2	2.5±0.2	0.3	1.0	1.4	12.5	×1.5	U(1kp)	L(4kp)
	16	10 μF	K/M	KGM32AR51C106□#	A3	3.2±0.3	2.5±0.2	2.5±0.2	0.3	1.0	1.4	5.0			

If there is a "" in the thickness code indicates the same thickness (T-dimension) but different L/W or P-dimension.

For test items, specifications, and measurement conditions, please refer to "Test Conditions and Standards".

High Dielectric Constant type : Standard Spec.1 : [High Dielectric Constant type(Class2) KGM/KGT Series (Standard Spec.1)]

Standard Spec.2 : [High Dielectric Constant type(Class2) KGM/KGT Series (Standard Spec.2)]

Load Life: Applied voltage in the load life test, expressed as a multiple of the rated voltage.

Please contact for capacitance value other than standard.

Multilayer Ceramic Chip Capacitors



General

KGM Series



Class2 : High Dielectric Constant type
X6S (EIA) [S6]

Part number list General KGM03 to 21 Series Temperature Characteristic: S6 Tolerance □: K: ±10%/ M: ±20%

■ Standard Spec .2

EIA Size (Series Code)	Voltage [Vdc]	Capacitance	Tolerance Code :□	Part Number	Thickness /tanδ Code	Dimension[mm]						tanδ max. [%]	Load Life	Packaging Code (QTY):#	
						L	W	T	P min.	P max.	P ~ P min.			φ180 Reel	φ330 Reel
0201 (KGM03)	10	1 μF	M	KGM03CS61A105M#	C10	0.6±0.09	0.3±0.09	0.3±0.09	0.13	0.23	0.19	20.0	×1.0	H(15kp)	N(50kp)
	6.3	1 μF	M	KGM03CS60J105M#											
	4	1 μF	M	KGM03CS60G105M#											
	2.5	4.7 μF	M	KGM03DS60E475MH	D9	0.6±0.09	0.3±0.09	0.5±0.05	0.13	0.23	0.19	15.0	×1.0	H(10kp)	— (—)
0402 (KGM05)	25	1 μF	K/M	KGM05AS61E105□#	A8	1.0±0.05	0.5±0.05	0.5±0.05	0.15	0.35	0.3	12.5	×1.0	H(10kp)	N(50kp)
	16	0.47 μF	M	KGM05AS61C474M#	A8	1.0±0.05	0.5±0.05	0.5±0.05	0.15	0.35	0.3	12.5	×1.0	H(10kp)	N(50kp)
		2.2 μF	K/M	KGM05AS61C225□#	A*8	1.0±0.2	0.5±0.2	0.55max.							
	10	0.47 μF	M	KGM05AS61A474M#	A8	1.0±0.05	0.5±0.05	0.5±0.05	0.15	0.35	0.3	12.5	×1.0	H(10kp)	N(50kp)
		1 μF	K/M	KGM05AS61A105□#	A8	1.0±0.05	0.5±0.05	0.5±0.05	0.15	0.35	0.3	12.5	×1.0	H(10kp)	N(50kp)
		4.7 μF	K/M	KGM05CS61A475□#	C8	1.0±0.2	0.5±0.2	0.5±0.2	0.15	0.35	0.3	12.5	×1.0	H(10kp)	N(40kp)
	6.3	4.7 μF	K/M	KGM05BS60J475□#	B8	1.0±0.15	0.5±0.15	0.5±0.15	0.15	0.35	0.3	12.5	×1.0	H(10kp)	N(40kp)
				KGM05CS60J475□#	C8	1.0±0.2	0.5±0.2	0.5±0.2	0.15	0.35	0.3	12.5	×1.5		
		10 μF	M	KGM05CS60J106M#	C8	1.0±0.2	0.5±0.2	0.5±0.2	0.15	0.35	0.3	12.5	×1.0	H(10kp)	N(40kp)
	4	22 μF	M	KGM05DS60G226M#	D8	1.0±0.2	0.5±0.2	0.8max.	0.15	0.35	0.3	12.5	×1.0	H(10kp)	N(30kp)
0603 (KGM15)	16	4.7 μF	K/M	KGM15CS61C475□#	C8	1.6±0.2	0.8±0.2	0.8±0.2	0.2	0.6	0.5	12.5	×1.0	T(4kp)	M(10kp)
		10 μF	M	KGM15CS61C106M#	C9	1.6±0.2	0.8±0.2	0.8±0.2	0.2	0.6	0.5	15.0			
	10	10 μF	K/M	KGM15CS61A106□#	C9	1.6±0.2	0.8±0.2	0.8±0.2	0.2	0.6	0.5	15.0	×1.5	T(4kp)	M(10kp)
		22 μF	M	KGM15CS61A226M#	C8	1.6±0.2	0.8±0.2	0.8±0.2	0.2	0.6	0.5	12.5	×1.0		
	6.3	22 μF	M	KGM15CS60J226M#	C8	1.6±0.2	0.8±0.2	0.8±0.2	0.2	0.6	0.5	12.5	×1.0	T(4kp)	M(10kp)
		22 μF	M	KGM15CS60G226M#	C8	1.6±0.2	0.8±0.2	0.8±0.2	0.2	0.6	0.5	12.5	×1.0	T(4kp)	M(10kp)
	4	47 μF	M	KGM15CS60G476M#	C8	1.6±0.2	0.8±0.2	0.8±0.2	0.2	0.6	0.5	12.5	×1.0		
	2.5	47 μF	M	KGM15CS60E476M#	C8	1.6±0.2	0.8±0.2	0.8±0.2	0.2	0.6	0.5	12.5	×1.0	T(4kp)	M(10kp)
		100 μF	M	KGM15CS60E107M#											
0805 (KGM21)	25	10 μF	M	KGM21KS61E106M#	K8	2.0±0.15	1.25±0.15	1.25±0.15		0.75	0.7	12.5	×1.0	U(3kp)	L(10kp)
		22 μF	M	KGM21AS61E226M#	A8	2.0±0.2	1.25±0.2	1.25±0.2							
	16	22 μF	M	KGM21AS61C226M#	A8	2.0±0.2	1.25±0.2	1.25±0.2	0.2	0.75	0.7	12.5	×1.0	U(3kp)	L(10kp)
	10	22 μF	M	KGM21AS61A226M#	A8	2.0±0.2	1.25±0.2	1.25±0.2	0.2	0.75	0.7	12.5	×1.5		
	6.3	22 μF	M	KGM21AS60J226M#	A8	2.0±0.2	1.25±0.2	1.25±0.2	0.2	0.75	0.7	12.5	×1.5	U(3kp)	L(10kp)
	4	47 μF	M	KGM21AS60G476M#	A7	2.0±0.2	1.25±0.2	1.25±0.2	0.2	0.75	0.7	10.0	×1.0	U(3kp)	L(10kp)
		100 μF	M	KGM21AS60G107M#	A8	2.0±0.2	1.25±0.2	1.25±0.2	0.2	0.75	0.7	12.5			

If there is a "" in the thickness code indicates the same thickness (T-dimension) but different L/W or P-dimension.

Class2 : High Dielectric Constant type
X6T (EIA) [T6]

Part number list General KGM03 Series Temperature Characteristics: T6 Tolerance □: K: ±10%/ M: ±20%

■ Standard Spec .2

EIA Size (Series Code)	Voltage [Vdc]	Capacitance	Tolerance Code :□	Part Number	Thickness /tanδ Code	Dimension[mm]						tanδ max. [%]	Load Life	Packaging Code (QTY):#	
						L	W	T	P min.	P max.	P ~ P min.			φ180 Reel	φ330 Reel
0201 (KGM03)	10	0.22 μF	K/M	KGM03CT61A224□#	C8	0.6±0.09	0.3±0.09	0.3±0.09	0.13	0.23	0.19	12.5	×1.0	H(15kp)	N(50kp)
	4	2.2 μF	M	KGM03CT60G225M#											
	2.5	1 μF	M	KGM03BT60E105M#	B8	0.6±0.05	0.3±0.05	0.3±0.05	0.13	0.23	0.19	12.5	×1.0	H(15kp) / Q(30kp)	N(50kp) / W(150kp)
		2.2 μF	M	KGM03CT60E225M#	C8	0.6±0.09	0.3±0.09	0.3±0.09	0.13	0.23	0.19	12.5	×1.5	H(15kp)	N(50kp)

For test items, specifications, and measurement conditions, please refer to "Test Conditions and Standards".
High Dielectric Constant type : Standard Spec.2 : [High Dielectric Constant type(Class2) KGM/KGT Series (Standard Spec.2)]

Load Life: Applied voltage in the load life test, expressed as a multiple of the rated voltage.

Please contact for capacitance value other than standard.

Multilayer Ceramic Chip Capacitors



General

KGM Series



Class2 : High Dielectric Constant type
X7R (EIA) [R7]

Part number list General KGM02 to 32 Series Temperature Characteristics: R7 Tolerance □: K: ±10%/ M: ±20%
■ Standard Spec.1 ■ Standard Spec.2

EIA Size (Series Code)	Voltage [Vdc]	Capacitance	Tolerance Code :□	Part Number	Thickness /tanδ Code	Dimension[mm]						tanδ max. [%]	Load Life	Packaging Code (QTY):#		
						L	W	T	P min.	P max.	P ~ P min.			φ180 Reel	φ330 Reel	
01005 (KGM02)	16	100 pF	K/M	KGM02AR71C101□#	A8	0.4±0.02	0.2±0.02	0.2±0.02	0.07	0.14	0.13	12.5	×1.5	H(20kp) / P(40kp)	N(80kp)	
		150 pF	K/M	KGM02AR71C151□#												
		220 pF	K/M	KGM02AR71C221□#												
		330 pF	K/M	KGM02AR71C331□#												
		470 pF	K/M	KGM02AR71C471□#	A8	0.4±0.02	0.2±0.02	0.2±0.02	0.07	0.14	0.13	12.5	×1.5	H(20kp) / P(40kp)	N(80kp)	
		680 pF	K/M	KGM02AR71C681□#												
		1 nF	K/M	KGM02AR71C102□#												
		1.5 nF	K/M	KGM02AR71C152□#												
2.2 nF	K/M	KGM02AR71C222□#														
0402 (KGM05)	25	0.1 μF	K/M	KGM05AR71E104□#	A8	1.0±0.05	0.5±0.05	0.5±0.05	0.15	0.35	0.3	12.5	×1.5	H(10kp)	N(50kp)	
		0.47 μF	K/M	KGM05AR70J474□#	A8											
		1 μF	K/M	KGM05AR70J105□#												
0603 (KGM15)	25	1 μF	K/M	KGM15AR71E105□#R000	A7	1.6±0.1	0.8±0.1	0.8±0.1	0.2	0.6	0.5	10.0	×1.5	T(4kp)	M(10kp)	
	10	2.2 μF	K/M	KGM15AR71A225□#												
0805 (KGM21)	50	1 μF	K/M	KGM21AR71H105□#	A3	2.0±0.2	1.25±0.2	1.25±0.2	0.2	0.75	0.7	5.0	×1.5	U(3kp)	L(10kp)	
	25	4.7 μF	K/M	KGM21AR71E475□#	A7	2.0±0.2	1.25±0.2	1.25±0.2	0.2	0.75	0.7	10.0				
	16	4.7 μF	K/M	KGM21KR71C475□#	K7	2.0±0.15	1.25±0.15	1.25±0.15	0.2	0.75	0.7	10.0				
		10 μF	K/M	KGM21AR71C106□#	A8	2.0±0.2	1.25±0.2	1.25±0.2	0.2	0.75	0.7	12.5				
	6.3	10 μF	K/M	KGM21AR70J106□#	A8	2.0±0.2	1.25±0.2	1.25±0.2	0.2	0.75	0.7	12.5				
1206 (KGM31)	50	4.7 μF	K/M	KGM31AR71H475□#	A3	3.2±0.2	1.6±0.2	1.6±0.2	0.3	0.85	1.4	5.0	×1.5	U(2.5kp)	L(5kp)	
		25	4.7 μF	K/M	KGM31AR71E475□#	A3	3.2±0.2	1.6±0.2	1.6±0.2	0.3	0.85	1.4				5.0
		10	10 μF	K/M	KGM31AR71E106□#	A3	3.2±0.2	1.6±0.2	1.6±0.2	0.3	0.85	1.4				5.0
		16	10 μF	K/M	KGM31AR71C106□#	A8	3.2±0.2	1.6±0.2	1.6±0.2	0.3	0.85	1.4				12.5
		10	22 μF	K/M	KGM31AR71A226□#	A5	3.2±0.2	1.6±0.2	1.6±0.2	0.3	0.85	1.4				7.5
		6.3	22 μF	K/M	KGM31AR70J226□#	A8	3.2±0.2	1.6±0.2	1.6±0.2	0.3	0.85	1.4				12.5
1210 (KGM32)	50	10 μF	K/M	KGM32AR71H106□#	A3	3.2±0.3	2.5±0.2	2.5±0.2	0.3	1.0	1.4	5.0	×1.5	U(1kp)	L(4kp)	
	25	10 μF	K/M	KGM32AR71E106□#	A8	3.2±0.3	2.5±0.2	2.5±0.2	0.3	1.0	1.4	12.5				

Class2 : High Dielectric Constant type
X7R (KAVX) [K7]

Part number list General KGM05 to 31 Series Temperature Characteristics: K7 Tolerance □: K: ±10%/ M: ±20%
■ Standard Spec.2

EIA Size (Series Code)	Voltage [Vdc]	Capacitance	Tolerance Code □	Part Number	Thickness /tanδ Code	Dimension[mm]						tanδ max. [%]	Load Life	Packaging Code (QTY):#	
						L	W	T	P min.	P max.	P ~ P min.			φ180 Reel	φ330 Reel
0402 (KGM05)	10	2.2 μF	K/M	KGM05BK71A225□#	B3	1.0±0.15	0.5±0.15	0.5±0.15	0.15	0.35	0.3	5.0	×1.0	H(10kp)	N(40kp)
0603 (KGM15)	16	4.7 μF	K/M	KGM15CK71C475□#	C8	1.6±0.2	0.8±0.2	0.8±0.2	0.2	0.6	0.5	12.5			
	10	10 μF	M	KGM15CK71A106M#	C9	1.6±0.2	0.8±0.2	0.8±0.2	0.2	0.6	0.5	15.0	×1.0	U(3kp)	L(10kp)
0805 (KGM21)	10	22 μF	M	KGM21AK71A226M#	A8	2.0±0.2	1.25±0.2	1.25±0.2	0.2	0.75	0.7	12.5			
1206 (KGM31)	100	2.2 μF	K/M	KGM31AK72A225□#	A3	3.2±0.2	1.6±0.2	1.6±0.2	0.3	0.85	1.4	5.0	×1.0	U(2.5kp)	L(5kp)
	4.7 μF	4.7 μF	K/M	KGM31HK72A475□U	H3	3.2±0.3	1.6±0.3	1.6±0.3	0.3	0.85	1.9	5.0			
	50	2.2 μF	K/M	KGM31AK71H225□#	A3	3.2±0.2	1.6±0.2	1.6±0.2	0.3	0.85	1.4	5.0			

For test items, specifications, and measurement conditions, please refer to "Test Conditions and Standards".

High Dielectric Constant type : Standard Spec.1 : [High Dielectric Constant type(Class2) KGM/KGT Series (Standard Spec.1)]
Standard Spec.2 : [High Dielectric Constant type(Class2) KGM/KGT Series (Standard Spec.2)]

Load Life: Applied voltage in the load life test, expressed as a multiple of the rated voltage.

Please contact for capacitance value other than standard.

Multilayer Ceramic Chip Capacitors

General

KGM Series



Class2 : High Dielectric Constant type
X7S (EIA) [S7]

Part number list General KGM03 to 31 Series Temperature Characteristics: S7 Tolerance □: K: ±10%/ M: ±20%
■ Standard Spec.1 ■ Standard Spec.2

EIA Size (Series Code)	Voltage [Vdc]	Capacitance	Tolerance Code :□	Part Number	Thickness /tanδ Code	Dimension[mm]						tanδ max. [%]	Load Life	Packaging Code (QTY):#	
						L	W	T	P min.	P max.	P ~ P min.			φ180 Reel	φ330 Reel
0201 (KGM03)	6.3	0.1 μF	K/M	KGM03AS70J104□#AX00	A7	0.6±0.03	0.3±0.03	0.3±0.03	0.1	0.2	0.2	10.0	×1.0	H(15kp) / Q(30kp)	N(50kp) / W(150kp)
0402 (KGM05)	10	2.2 μF	K/M	KGM05BS71A225□#	B3	1.0±0.15	0.5±0.15	0.5±0.15	0.15	0.35	0.3	5.0	×1.0	H(10kp)	N(40kp)
	6.3	1 μF	K/M	KGM05AS70J105□#	A8	1.0±0.05	0.5±0.05	0.5±0.05	0.15	0.35	0.3	12.5	×1.5	H(10kp)	N(50kp)
		2.2 μF	K/M	KGM05BS70J225□#	B3	1.0±0.15	0.5±0.15	0.5±0.15	0.15	0.35	0.3	5.0	×1.0	H(10kp)	N(40kp)
	4	1 μF	K/M	KGM05AS70G105□#	A8	1.0±0.05	0.5±0.05	0.5±0.05	0.15	0.35	0.3	12.5	×1.0	H(10kp)	N(50kp)
		2.2 μF	M	KGM05BS70G225M#	B3	1.0±0.15	0.5±0.15	0.5±0.15	0.15	0.35	0.3	5.0	×1.0	H(10kp)	N(40kp)
0805 (KGM21)	100	1 μF	K/M	KGM21AS72A105□#	A3	2.0±0.2	1.25±0.2	1.25±0.2	0.2	0.75	0.7	5.0	×1.0	U(3kp)	L(10kp)
1206 (KGM31)	100	2.2 μF	K/M	KGM31AS72A225□#	A3	3.2±0.2	1.6±0.2	1.6±0.2	0.3	0.85	1.4	5.0	×1.0	U(2.5kp)	L(5kp)
		4.7 μF	K/M	KGM31HS72A475□U	H3	3.2±0.3	1.6±0.3	1.6±0.3	0.3	0.85	1.9	5.0	×1.0	U(2kp)	—(—)

Class2 : High Dielectric Constant type
X7T (EIA) [T7]

Part number list General KGM03 to 21 Series Temperature Characteristics: T7 Tolerance □: K: ±10%/ M: ±20%
■ Standard Spec.2

EIA Size (Series Code)	Voltage [Vdc]	Capacitance	Tolerance Code :□	Part Number	Thickness /tanδ Code	Dimension[mm]						tanδ max. [%]	Load Life	Packaging Code (QTY):#	
						L	W	T	P min.	P max.	P ~ P min.			φ180 Reel	φ330 Reel
0201 (KGM03)	6.3	1 μF	M	KGM03CT70J105M#	C8	0.6±0.09	0.3±0.09	0.3±0.09	0.13	0.23	0.19	12.5	×1.0	H(15kp)	N(50kp)
0402 (KGM05)	10	4.7 μF	K/M	KGM05CT71A475□#	C8	1.0±0.2	0.5±0.2	0.5±0.2	0.15	0.35	0.3	12.5	×1.0	H(10kp)	N(40kp)
0603 (KGM15)	16	4.7 μF	K/M	KGM15CT71C475□#	C8	1.6±0.2	0.8±0.2	0.8±0.2	0.2	0.6	0.5	12.5	×1.0	T(4kp)	M(10kp)
	10	4.7 μF	K/M	KGM15CT71A475□#	C8	1.6±0.2	0.8±0.2	0.8±0.2	0.2	0.6	0.5	12.5	×1.5		
		10 μF	M	KGM15CT71A106M#	C9	1.6±0.2	0.8±0.2	0.8±0.2	0.2	0.6	0.5	15.0	×1.0		
	6.3	4.7 μF	K/M	KGM15CT70J475□#	C8	1.6±0.2	0.8±0.2	0.8±0.2	0.2	0.6	0.5	12.5	×1.0		
		10 μF	M	KGM15CT70J106M#	C9	1.6±0.2	0.8±0.2	0.8±0.2	0.2	0.6	0.5	15.0	×1.5		
0805 (KGM21)	10	22 μF	M	KGM21AT71A226M#	A8	2.0±0.2	1.25±0.2	1.25±0.2	0.2	0.75	0.7	12.5	×1.0	U(3kp)	L(10kp)
	6.3	22 μF	M	KGM21AT70J226M#	A8	2.0±0.2	1.25±0.2	1.25±0.2	0.2	0.75	0.7	12.5	×1.5		

For test items, specifications, and measurement conditions, please refer to "Test Conditions and Standards".

High Dielectric Constant type : Standard Spec.1 : [High Dielectric Constant type(Class2) KGM/KGT Series (Standard Spec.1)]
Standard Spec.2 : [High Dielectric Constant type(Class2) KGM/KGT Series (Standard Spec.2)]

Load Life: Applied voltage in the load life test, expressed as a multiple of the rated voltage.

Please contact for capacitance value other than standard.

Multilayer Ceramic Chip Capacitors

Test Conditions and Standards

Specifications and Test Conditions for Temperature-Compensating type (Class1)
KGM/KGU Series (■ Standard Spec.1) CG / CH

Test Items		Specifications	Test Conditions (Complies with JIS C5101/ IEC60384)		
Capacitance		Within tolerance	Capacitance	Measuring Frequency	Measuring Voltage
Q		C≥30pF : Q≥1000	C≤1000pF	1MHz ±10%	0.5 to 5Vrms
		C<30pF : Q≥400+20C	C>1000pF	1kHz ±10%	
Insulation Resistance		Over 10000MΩ or 500MΩ・μF, whichever is less	Measure after applying rated voltage for 1 minute in normal temperature and humidity. The charge and discharge current of the capacitor must not exceed 50mA.		
Dielectric Strength		Resist without problem	Apply*3 times of the rated voltage for 1 to 5 seconds. *KGU02ACΔ1ER20-220 : twice of the rated voltage for 1 to 5 seconds. The charge and discharge current of the capacitor must not exceed 50mA.		
Appearance		No problem observed	Microscope		
Termination Strength		No problem observed	Apply a sideward force of 1N to a PCB-mounted sample.		
Bending Strength		No significant damage with 1mm bending.	Glass epoxy PCB (90mm fulcrum spacing), 10 seconds duration time.		
Vibration	Appearance	No problem observed	Vibration frequency: 10 to 55 (Hz)		
	Capacitance	Within Tolerance	Amplitude: 1.5mm		
	Q	C≥30pF : Q≥1000 C<30pF : Q≥400+20C	Sweeping condition: 10→55→10Hz/ 1 minute in X, Y and Z directions: 2 hours each, 6 hours in total.		
Resistance to Solder Heat	Appearance	No problem observed	Soak the sample in 260±5°C solder for 10±0.5 seconds, and measure after resting in normal temperature and humidity for 24±2 hours.		
	Capacitance Variation	Within±2.5% or±0.25pF, whichever is larger	(Pre-heating conditions before soak)		
	Q	C≥30pF : Q≥1000 C<30pF : Q≥400+20C	Order	Temperature	Time
	Insulation Resistance	Over 10000MΩ or 500MΩ・μF whichever is less	1	80 to 100°C	2 minutes
			2	150 to 200°C	2 minutes
Dielectric strength	Resist without problem	The charge and discharge current of the capacitor must not exceed 50mA for IR and dielectric strength measurement.			
Solderability		Solder coverage : 95% min.	Soaking condition		
			Solder Type	Temperature	Time
			Sn-3.0Ag-0.5Cu	245±5°C	3±0.5 sec.
Temperature Cycle	Appearance	No problem observed	(Cycle)		
	Capacitance Variation	Within±2.5% or ±0.25pF, whichever is larger	Lowest operation temperature (30 min.)→ Room temperature (3 min.)→		
	Q	C≥30pF : Q≥1000 C<30pF : Q≥400+20C	Highest operation temperature (30 min.)→ Room temperature (3 min.)		
	Insulation Resistance	Over 10000MΩ or 500MΩ・μF, whichever is less	After 5 cycles, measure after 24±2 hours.		
	Dielectric Strength	Resist without problem	The charge and discharge current of the capacitor must not exceed 50mA for IR and dielectric strength measurement.		
Load Humidity	Appearance	No problem observed	Apply the rated voltage for 500+12/ -0 hours in the condition of 40±2°C and 90 to 95%RH, and measure after resting in normal temperature and humidity for 24±2 hours.		
	Capacitance Variation	Within±7.5% or ±0.75pF, whichever is larger	The charge and discharge current of the capacitor must not exceed 50mA for IR measurement.		
	Q	C≥30pF : Q≥200 C<30pF : Q≥100+10C/ 3			
	Insulation Resistance	Over 500MΩ or 25MΩ・μF, whichever is less			
Load Life	Appearance	No problem observed.	Apply *twice the rated voltage in 125±3°C for 1000+12/ -0 hours, and measure after resting in normal temperature and humidity for 24±2 hours.		
	Capacitance Variation	Within ±3% or ±0.3pF, whichever is larger	The charge and discharge current of the capacitor must not exceed 50mA for IR measurement.		
	Q	C≥30pF : Q≥350 10pF<C<30pF : Q≥275+5C/ 2 C<10pF : Q≥200+10C	*Products listed below shall apply each indicated voltage.		
	Insulation Resistance	Over 1000MΩ or 50MΩ・μF, whichever is less			

Voltage to be applied in the Load Life (Applied voltage is the multiple of the rated voltage)

Applied Voltage	Series	Dielectric	Rated Voltage	Products
$\times 1.2$	KGU02	CG, CH	25Vdc	KGU02ACG1ER20-220, KGU02ACH1ER20-220

Multilayer Ceramic Chip Capacitors

Test Conditions and Standards

Specifications and Test Conditions for High Dielectric Constant type (Class2)
KGM/KGT Series (■ Standard Spec.1) X5R [R5] / X7R [R7] / X7S [S7]

Test Items		Specifications	Test Conditions (Complies with JIS C5101/ IEC60384)		
Capacitance		Within tolerance	Measure after heat treatment		
			Capacitance	Measuring Frequency	Measuring Voltage
tanδ		Refer to part number list	C≤10μF	1kHz ±10%	1.0±0.2Vrms
			C>10μF	120Hz ±10%	*0.5±0.1Vrms
			*KGM02AR50J104 The charge and discharge current of the capacitor must not exceed 50mA.		
Insulation Resistance		Over 10000MΩ or 500MΩ・μF, whichever is less	Measure after applying rated voltage for 1 minute in normal temperature and humidity. The charge and discharge current of the capacitor must not exceed 50mA.		
Dielectric Strength		Resist without problem	Apply *2.5 times of the rated voltage for 1 to 5 seconds. *KGM31AR52A225, KGM31AS72A225 : twice of the rated voltage for 1 to 5 seconds. The charge and discharge current of the capacitor must not exceed 50mA.		
Appearance		No problem observed	Microscope		
Termination Strength		No problem observed	Apply a sideward force of 5N to a PCB-mounted sample. note : 2N for 0201 size, and 1N for 01005 size. Exclude KGT series.		
Bending Strength		No significant damage with 1mm bending	Glass epoxy PCB (90mm fulcrum spacing), 10 seconds duration time. Exclude KGT series.		
Vibration	Appearance	No problem observed	Take the initial value after heat treatment. Vibration frequency: 10 to 55 (Hz)		
	Capacitance	Within tolerance	Amplitude: 1.5mm		
	tanδ	Within tolerance	Sweeping condition: 10→55→10Hz/ 1 minute in X, Y and Z directions: 2 hours each, 6 hours in total, then measure the sample after heat treatment.		
Resistance to Solder Heat	Appearance	No problem observed	Take the initial value after heat treatment.		
	Capacitance Variation	Within±7.5%	Soak the sample in 260±5°C solder for 10±0.5 seconds, and measure after heat treatment.		
	tanδ	Within tolerance	(Pre-heating conditions before soak)		
	Insulation Resistance	Over 10000MΩ or 500MΩ・μF, whichever is less	Order	Temperature	Time
	Dielectric Strength	Resist without problem	1	80 to 100°C	2 minutes
			2	150 to 200°C	2 minutes
			The charge and discharge current of the capacitor must not exceed 50mA for IR and dielectric strength measurement.		
Solderability		Solder coverage : 95% min.	Soaking condition		
			Solder Type	Temperature	Time
			Sn-3.0Ag-0.5Cu	245±5°C	3±0.5 sec.

Multilayer Ceramic Chip Capacitors

Test Conditions and Standards

Specifications and Test Conditions for High Dielectric Constant type (Class2)
KGM/KGT Series (■ Standard Spec.1) X5R [R5] / X7R [R7] / X7S [S7]

Test Items		Specifications	Test Conditions (Complies with JIS C5101/ IEC60384)
Temperature Cycle	Appearance	No problem observed	Take the initial value after heat treatment.
	Capacitance Variation	Within±7.5%	(Cycle) Lowest operation temperature (30 min.)→
	tanδ	Within tolerance	Room temperature (3 min.)→
	Insulation Resistance	Over 10000MΩ or 500MΩ・μF, whichever is less	Highest operation temperature (30 min.)→ Room temperature (3 min.) After 5 cycles, measure after heat treatment.
	Dielectric Strength	Resist without problem	The charge and discharge current of the capacitor must not exceed 50mA for IR and dielectric strength measurement.
Load Humidity	Appearance	No problem observed	Take the initial value after heat treatment.
	Capacitance Variation	Within±12.5%	Apply rated voltage for 500+12/-0 hours in 40±2°C and 90 to 95%RH, and measure after heat treatment.
	tanδ	200% max. of initial value	The charge and discharge current of the capacitor must not exceed 50mA for IR measurement.
	Insulation Resistance	Over 500MΩ or 25MΩ・μF, whichever is less	
Load Life	Appearance	No problem observed	Take the initial value after heat treatment.
	Capacitance Variation	Within±12.5%	Apply *1.5 times the rated voltage at the highest operation temperature for 1000+12/-0 hours, and measure after heat treatment.
	tanδ	200% max. of initial value	The charge and discharge current of the capacitor must not exceed 50mA for IR measurement.
	Insulation Resistance	Over 1000MΩ or 50MΩ・μF, whichever is less	*Products listed below shall apply each indicated voltage.
Heat treatment		Expose sample in the temperature of 150+0/-10°C for 1 hour and leave the sample in normal temperature and humidity for 24±2 hours.	

Voltage to be applied in the Load Life (Applied voltage is the multiple of the rated voltage)

Applied Voltage	Series	Dielectric	Rated Voltage	Products
×1.0	KGM02	R5	10 Vdc	KGM02AR51A104
	KGM21	R7	16 Vdc	KGM21AR71C106
	KGM31	R5	100 Vdc	KGM31AR52A225
		S7	100 Vdc	KGM31AS72A225
×1.3	KGM02	R5	6.3 Vdc	KGM02AR50J153-104
	KGT03	R5	6.3 Vdc	KGT03YR50J104

Multilayer Ceramic Chip Capacitors

Test Conditions and Standards

Specifications and Test Conditions for High Dielectric Constant type (Class2)

KGM/KGT Series (■ Standard Spec.2) X5R [R5] / X6S [S6] / X6T [T6] / X7R [R7,K7] / X7S [S7] / X7T [T7]

Test Items		Specifications	Test Conditions (Complies with JIS C5101/ IEC60384)		
Capacitance		Within tolerance	Measure after heat treatment		
			Capacitance	Measuring Frequency	Measuring Voltage
			C≤10μF	1kHz ±10%	1.0±0.2Vrms
			C>10μF	120Hz ±10%	*0.5±0.1Vrms
tanδ		Refer to part number list	The table below lists the applicable product numbers for measurement voltages marked with an asterisk (*). The charge and discharge current of the capacitor must not exceed 50mA.		
Insulation Resistance		Over 50MΩ・μF	Measure after applying rated voltage for 1 minute in normal temperature and humidity. The charge and discharge current of the capacitor must not exceed 50mA.		
Dielectric Strength		Resist without problem	Apply *2.5 times of the rated voltage for 1 to 5 seconds. *KGM21AS72A105, KGM31HS72A475 KGM31AK72A225, KGM31HK72A475 : twice of the rated voltage for 1 to 5 seconds. The charge and discharge current of the capacitor must not exceed 50mA.		
Appearance		No problem observed	Microscope		
Termination Strength		No problem observed	Apply a sideward force of 5N to a PCB-mounted sample. note : 2N for 0201 size, and 1N for 01005 size. Exclude KGT series.		
Bending Strength		No significant damage with 1mm bending	Glass epoxy PCB (90mm fulcrum spacing), 10 seconds duration time. Exclude KGT series.		
Vibration	Appearance	No problem observed	Take the initial value after heat treatment.		
	Capacitance	Within tolerance	Vibration frequency: 10 to 55 (Hz) Amplitude: 1.5mm		
	tanδ	Within tolerance	Sweeping condition: 10→55→10Hz/ 1 minute in X, Y and Z directions: 2 hours each, 6 hours in total, then measure the sample after heat treatment.		
Resistance to Solder Heat	Appearance	No problem observed	Take the initial value after heat treatment.		
	Capacitance Variation	Within±7.5%	Soak the sample in 260±5°C solder for 10±0.5 seconds and measure after heat treatment.		
	tanδ	Within tolerance	(Pre-heating conditions before soak)		
	Insulation Resistance	Over 50MΩ・μF	Order	Temperature	Time
	Dielectric Strength	Resist without problem	1	80 to 100°C	2 minutes
			2	150 to 200°C	2 minutes
			The charge and discharge current of the capacitor must not exceed 50mA for IR and dielectric strength measurement.		
Solderability		Solder coverage : 95% min.	Soaking condition		
			Solder Type	Temperature	Time
			Sn-3.0Ag-0.5Cu	245±5°C	3±0.5 sec.

Applicable Product Numbers for Measurement Voltage 0.5 $\pm$ 0.1Vrms

Series	Dielectric	Rated Voltage	Products
KGM02	R5	6.3 Vdc	KGM02AR50J474
KGM03	R5	6.3 Vdc	KGM03CR50J225, KGM03BR50J225, KGM03DR50J475
		4 Vdc	KGM03CR50G475, KGM03DR50G106
	S6	2.5 Vdc	KGM03DS60E475
	T6	4 Vdc	KGM03CT60G225
		2.5 Vdc	KGM03BT60E105, KGM03CT60E225
	S7	6.3 Vdc	KGM03AS70J104□#AX00
KGM05	R5	6.3 Vdc	KGM05CR50J106
	S6	6.3 Vdc	KGM05CS60J106
	R7	6.3 Vdc	KGM05AR70J474
KGT03	R5	6.3 Vdc	KGT03YR50J105
	T6	4 Vdc	KGT03YT60G105
KGT05	R5	6.3 Vdc	KGT05ZR50J106, KGT05YR50J475
	T6	4 Vdc	KGT05XT60G105
		2.5 Vdc	KGT05ZT60E106

Multilayer Ceramic Chip Capacitors Test Conditions and Standards

Specifications and Test Conditions for High Dielectric Constant type (Class2)

KGM/KGT Series (■ Standard Spec.2) X5R [R5] / X6S [S6] / X6T [T6] / X7R [R7,K7] / X7S [S7] / X7T [T7]

Test Items		Specifications	Test Conditions (Complies with JIS C5101/ IEC60384)
Temperature Cycle	Appearance	No problem observed	Take the initial value after heat treatment. (Cycle) Lowest operation temperature (30 min.)→ Room temperature (3 min.)→ Highest operation temperature (30 min.)→ Room temperature (3 min.) After 5 cycles, measure after heat treatment. The charge and discharge current of the capacitor must not exceed 50mA for IR and dielectric strength measurement.
	Capacitance Variation	Within±7.5%	
	tanδ	Within tolerance	
	Insulation Resistance	Over 50MΩ・μF	
	Dielectric Strength	Resist without problem	
Load Humidity	Appearance	No problem observed	Take the initial value after heat treatment. Apply rated voltage for 500+12/ -0 hours in 40±2℃ and 90 to 95%RH, and measure after heat treatment. The charge and discharge current of the capacitor must not exceed 50mA for IR measurement.
	Capacitance Variation	Within±12.5%	
	tanδ	200% max. of initial value	
	Insulation Resistance	Over 10MΩ・μF	
Load Life	Appearance	No problem observed	Take the initial value after heat treatment. Apply *One time the rated voltage at the highest operation temperature for 1000+12/ -0 hours, and measure after heat treatment. The charge and discharge current of the capacitor must not exceed 50mA for IR measurement. *Products listed below shall apply each indicated voltage.
	Capacitance Variation	Within±12.5%	
	tanδ	200% max. of initial value	
	Insulation Resistance	Over 10MΩ・μF	
Heat treatment		Expose sample in the temperature of 150+0/ -10℃ for 1 hour and leave the sample in normal temperature and humidity for 24±2 hours.	

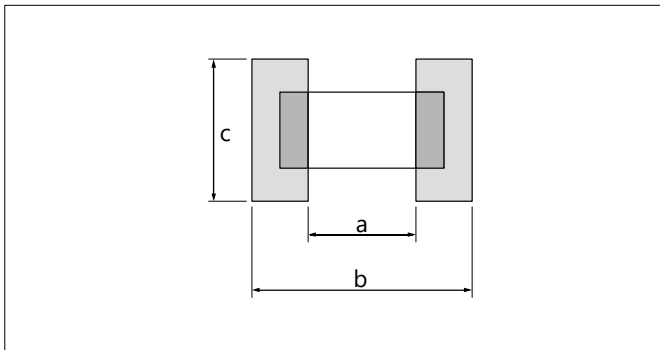
Voltage to be applied in the Load Life (Applied voltage is the multiple of the rated voltage)

Applied Voltage	Series	Dielectric	Rated Voltage	Products
×1.2	KGM03	R5	6.3 Vdc	KGM03BR50J105
×1.3	KGM03	R5	10 Vdc	KGM03AR51A223-224
			6.3 Vdc	KGM03AR50J474
×1.5	KGM03	R5	10 Vdc	KGM03CR51A105
		T6	2.5 Vdc	KGM03CT60E225
	KGM05	R5	16 Vdc	KGM05AR51C105
			10 Vdc	KGM05AR51A474-225, KGM05CR51A475
			6.3 Vdc	KGM05AR50J225
		S6	6.3 Vdc	KGM05CS60J475
		R7	6.3 Vdc	KGM05AR70J474, KGM05AR70J105
		S7	6.3 Vdc	KGM05AS70J105
	KGM15	R5	16 Vdc	KGM15CR51C106
		S6	10 Vdc	KGM15CS61A106
		R7	25 Vdc	KGM15AR71E105□#R000
			10 Vdc	KGM15AR71A225
		T7	10 Vdc	KGM15CT71A475
			6.3 Vdc	KGM15CT70J106
	KGM21	R5	10 Vdc	KGM21AR51A226
		S6	6.3 Vdc	KGM21AS60J226
		R7	25 Vdc	KGM21AR71E475
			16 Vdc	KGM21KR71C475
		T7	6.3 Vdc	KGM21AT70J226
	KGM31	R7	50 Vdc	KGM31AR71H475
			25 Vdc	KGM31AR71E106
	KGT05	T6	2.5 Vdc	KGT05ZT60E106
×2.0	KGM31	R7	25 Vdc	KGM31AR71E475

Multilayer Ceramic Chip Capacitors Test Conditions and Standards

Substrate for Adhesion Strength Test, Vibration Test, Soldering Heat Resistance Test, Temperature Cycle Test, Load Humidity Test, High-Temperature with Loading Test.

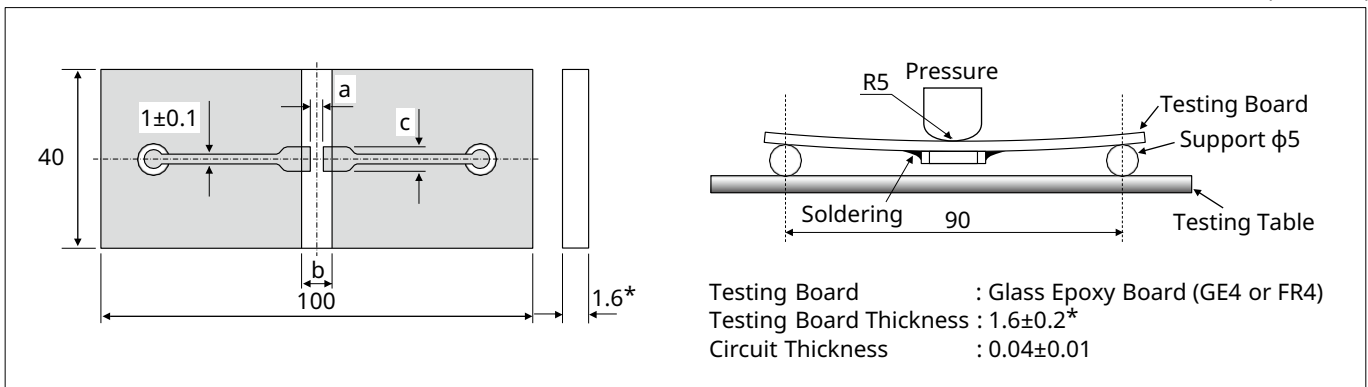
(Unit: mm)



EIA Size (Size Code)	a	b	c
01005 (02)	0.15	0.5	0.2
0201 (03)	0.26	0.92	0.32
0402 (05)	0.4	1.4	0.5
0603 (15)	1.0	3.0	1.2
0805 (21)	1.2	4.0	1.65
1206 (31)	2.2	5.0	2.0
1210 (32)	2.2	5.0	2.9

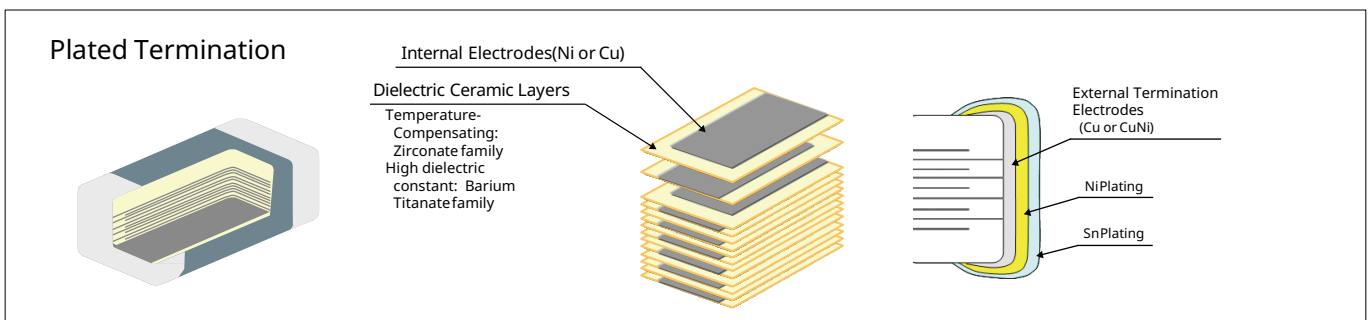
Substrate for Bending Test

(Unit: mm)



* 02, 03, 05 size 0.8±0.1mm

Structure



■ Certification status

<ISO>

Acquired ISO 9001 quality management system certification.

<IATF>

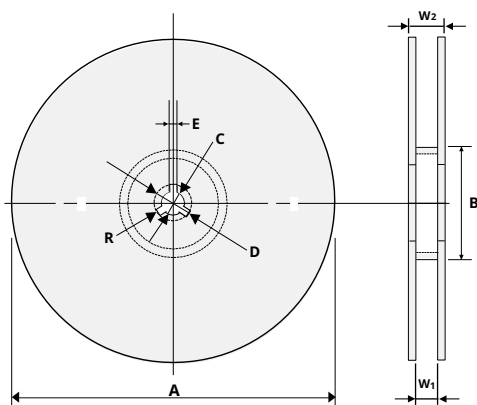
Acquired IATF 16949 quality management system certification.

■ Production plant

Kagoshima Kokubu plant

Multilayer Ceramic Chip Capacitors Packaging Options Tape and Reel

Reel

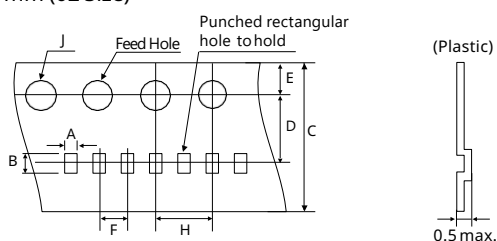


(Unit: mm)

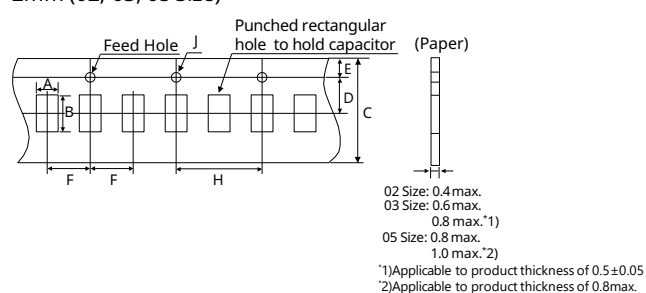
Code Reel	A	B	C	D
φ180 Reel (Code: T, H, Q, U)	180 ⁺⁰ _{-2.0}	φ60 min.	13 ±0.5	21±0.8
φ180 Reel (Code: P)	178 ±2.0			
φ330 Reel (Code: L, M, N, W)	330 ±2.0			
Code Reel	E	W ₁	W ₂	R
φ180 Reel (Code: T, H, Q, U)	2.0 ±0.5	10.5 ±1.5	16.5 max.	1.0
φ180 Reel (Code: P)		4.35 ±0.3	6.95 ±1.0	
φ330 Reel (Code: L, M, N, W)		9.5 ±1.0	16.5 max.	

Carrier Tape

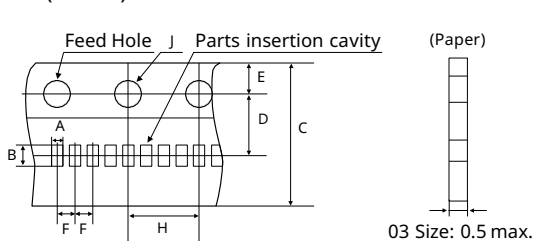
F = 1mm (02 Size)



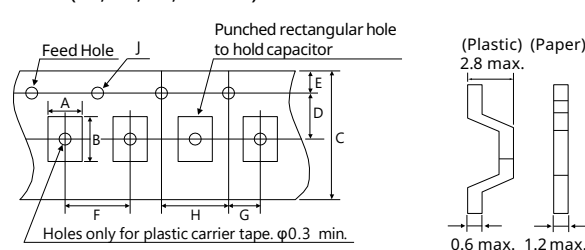
F = 2mm (02, 03, 05 Size)



F = 1mm (03 Size)



F = 4mm (15, 21, 31, 32 Size)



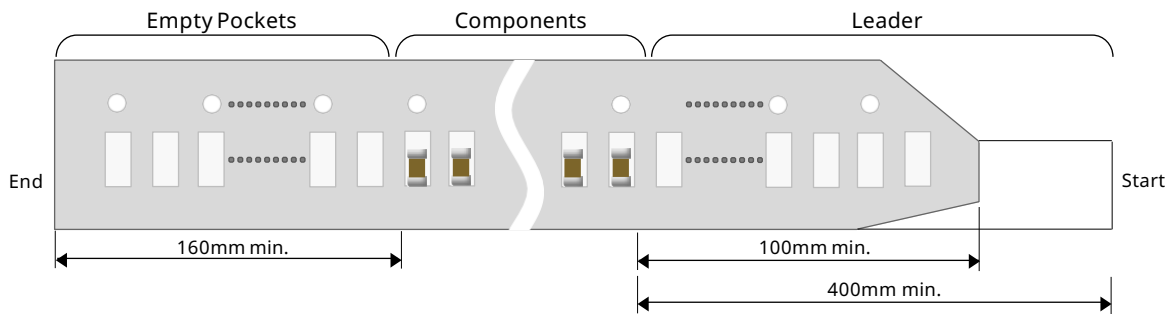
(Unit: mm)

EIA Size (Size Code)	A	B	C	D	E	F	G	H	J	Carrier Tape		
										Width	Material	
01005 (02)*	0.24 ±0.02	0.44 ±0.02	4.0 ±0.08	1.8 ±0.02	0.9 ±0.05	1.0 ±0.02	-	2.0 ±0.04	0.8±0.04	4	Plastic	
	0.25 ±0.03	0.45 ±0.03	8.0 ±0.3	3.5 ±0.05	1.75 ±0.1	2.0 ±0.05		4.0 ±0.1	1.5±0.1/-0	8	Paper	
0201 (03)*	0.37 ±0.03	0.67 ±0.03	8.0+0.3/-0.1	3.5 ±0.05	1.75 ±0.1	1.0 ±0.05	-	4.0 ±0.05	1.5±0.1/-0	8	Paper	
			8.0 ±0.3			2.0 ±0.05		4.0 ±0.1				
	0.39 ±0.03	0.69 ±0.03	8.0 ±0.3	3.5 ±0.05	1.75 ±0.1	2.0 ±0.05		4.0 ±0.1	1.5±0.1/-0			
	0.42 ±0.03	0.72 ±0.03										
	0.44 ±0.05	0.74 ±0.05										
0402 (05)*	0.65 ±0.1	1.15 ±0.1	8.0 ±0.3	3.5 ±0.05	1.75 ±0.1	2.0 ±0.05	-	4.0 ±0.1	1.5±0.1/-0	8	Paper	
0.75 ±0.1												
0.8 ±0.1	1.3 ±0.1											
0603 (15)*	1.0 ±0.2	1.8 ±0.2	8.0 ±0.3	3.5 ±0.05	1.75 ±0.1	4.0 ±0.1	2.0 ±0.05	4.0 ±0.1	1.5±0.1/-0	8	Paper	
	1.1 ±0.2	1.9 ±0.2										
0805 (21)	1.5 ±0.2	2.3 ±0.2	8.0 ±0.3	3.5 ±0.05	1.75 ±0.1	4.0 ±0.1	2.0 ±0.05	4.0 ±0.1	1.5±0.1/-0	8	Paper Plastic	
1206 (31)	2.0 ±0.2	3.6 ±0.2	8.0 ±0.3	3.5 ±0.05	1.75 ±0.1	4.0 ±0.1	2.0 ±0.05	4.0 ±0.1	1.5±0.1/-0	8	Paper Plastic	
1210 (32)	2.9 ±0.2	3.6 ±0.2	8.0 ±0.3	3.5 ±0.05	1.75 ±0.1	4.0 ±0.1	2.0 ±0.05	4.0 ±0.1	1.5±0.1/-0	8	Plastic	

* Option

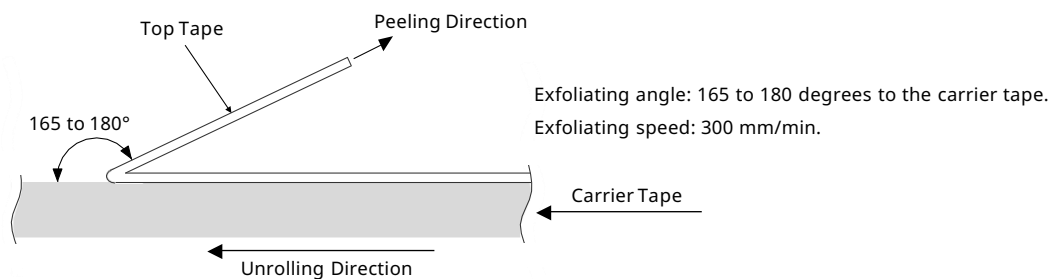
Multilayer Ceramic Chip Capacitors Packaging Options

Detail of leader and trailer



Adhesive tape

- 1) The exfoliative strength when peeling off the top tape from the carrier tape by the method of the following figure shall be *0.1 to 0.7N. *02 Size: 0.1 to 0.5N
- 2) When the top tape is peeled off, the adhesive stays on the top tape.
- 3) Chip capacitors will be in a state free without being stuck on the thermal adhesive tape.



Carrier tape

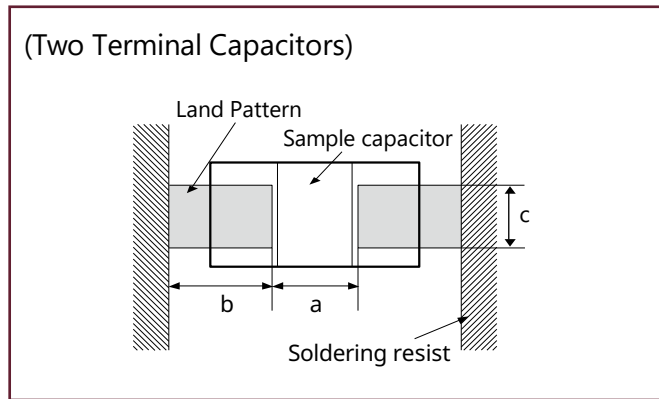
- 1) Chip will not fall off from carrier tape or carrier tape will not be damaged by bending than within a radius of 25mm.
- 2) The chip are inserted continuously without any empty pocket.
- 3) Chip will not be mis-mounted because of too big clearance between components and cavity. Also the waste of carrier tape will not fill a nozzle hole of mounting machine.

Multilayer Ceramic Chip Capacitors Surface Mounting Information

Dimensions for recommended typical land

Since the amount of solder (size of fillet) to be used has direct influence on the capacitor after mounting, the sufficient consideration is necessary.

When the amounts of solder is too much, the stress that a capacitor receives becomes larger. It may become the cause of a crack in the capacitor. When the land design of printed wiring board is considered, it is necessary to set up the form and size of land pattern so that the amount of solder is suitable.



Two Terminal Capacitors

(Unit: mm)

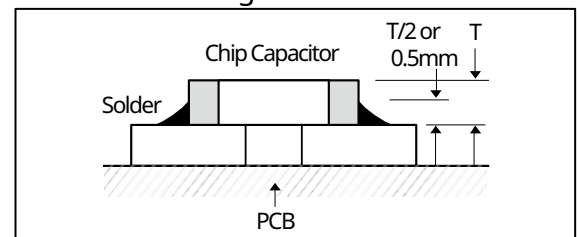
EIA Size (Size Code)	Dimension		Recommended land dimensions		
	L	W	a	b	c
01005 (02)	0.4 ±0.02	0.2 ±0.02	0.13 to 0.2	0.12 to 0.18	0.2 to 0.23
0201 (03)	0.6 ±0.03	0.3 ±0.03	0.2 to 0.25	0.25 to 0.35	0.3 to 0.4
	0.6 ±0.05	0.3 ±0.05			
	0.6 ±0.09	0.3 ±0.09	0.23 to 0.3	0.25 to 0.35	0.3 to 0.45
0402 (05)	1.0 ±0.05	0.5 ±0.05	0.3 to 0.5	0.35 to 0.45	0.4 to 0.6
	1.0 ±0.15	0.5 ±0.15			
	1.0 ±0.2	0.5 ±0.2	0.4 to 0.6	0.4 to 0.5	0.5 to 0.75
0603 (15)	1.6 ±0.1	0.8 ±0.1	0.7 to 1.0	0.8 to 1.0	0.6 to 0.9
	1.6 ±0.2	0.8 ±0.2	0.8 to 1.0	0.8 to 1.0	0.8 to 1.1
0805 (21)	2.0 ±0.2	1.25 ±0.2	1.0 to 1.3	1.0 to 1.2	1.25 to 1.55
1206 (31)	3.2 ±0.2	1.6 ±0.15	2.1 to 2.5	1.1 to 1.3	1.4 to 1.9
	3.2 ±0.2	1.6 ±0.2			
	3.2 ±0.3	1.6 ±0.3	2.1 to 2.5	1.1 to 1.3	1.6 to 2.0
1210 (32)	3.2 ±0.3	2.5 ±0.2	2.1 to 2.5	1.1 to 1.3	1.9 to 2.8

* Recommended land dimensions may differ depending on dimensional tolerance.

Design of printed circuit and Soldering

The recommended fillet height shall be 1/2 of the thickness of capacitors or 0.5mm. When mounting two or more capacitors in the common land, it is necessary to separate the land with the solder resist strike so that it may become the exclusive land of each capacitor.

Ideal Solder Height

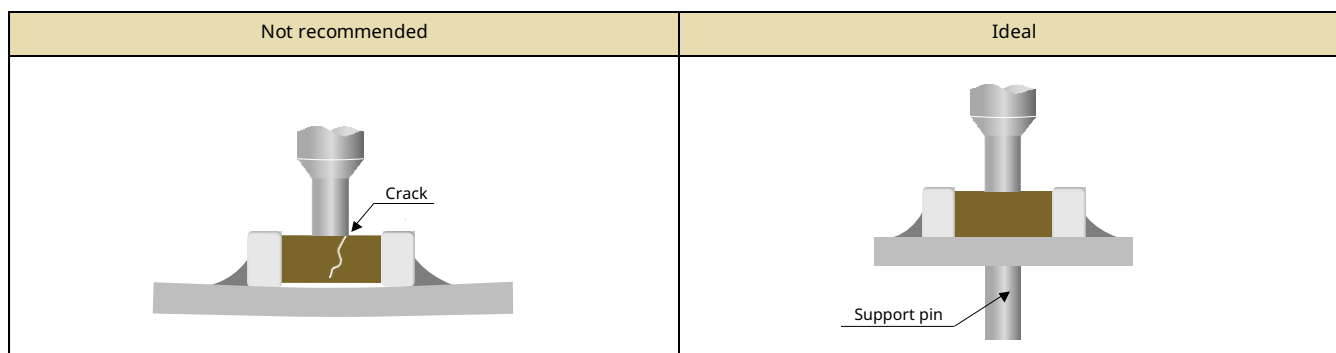


Item	Prohibited	Recommended example : Separation by solder resist
Multiple parts mount		
Mount with leaded parts		
Wire soldering after mounting		
Side by side layout		

Multilayer Ceramic Chip Capacitors Surface Mounting Information

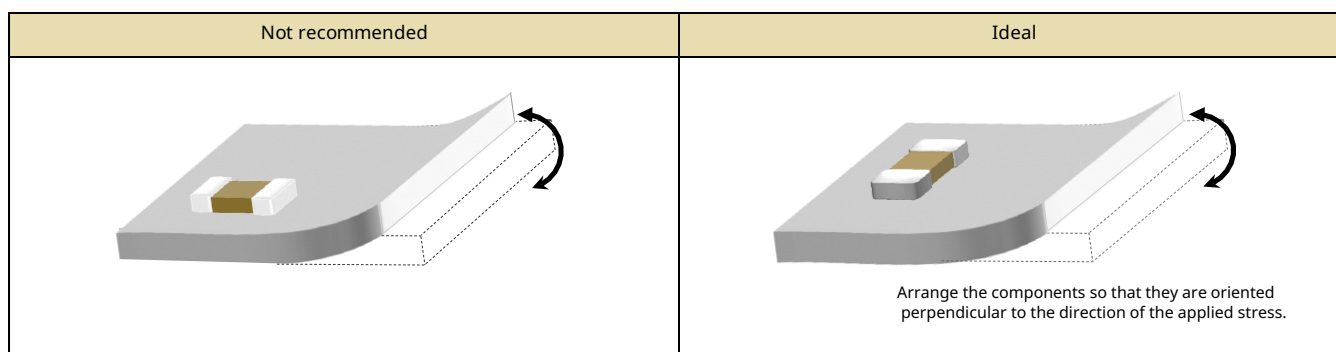
Actual Mounting

- 1) If the position of the vacuum nozzle is too low, a large force may be applied to the chip capacitor during mounting, resulting in cracking.
- 2) During mounting, set the nozzle pressure to a static load of 1 to 3 N.
- 3) To minimize the shock of the vacuum nozzle, provide a support pin on the back of the PCB to minimize PCB flexure.
- 4) Bottom position of pick up nozzle should be adjusted to the top surface of a substrate which camber is corrected.



Mounting Design

After soldering capacitors to the PCB, bending of the board during subsequent processes — such as board cutting, breaking, inspection, component mounting, chassis assembly, or flow soldering of the reverse side after reflow — or during handling, may cause chip cracking. To prevent damage, position the capacitors to minimize stress from PCB flexure.

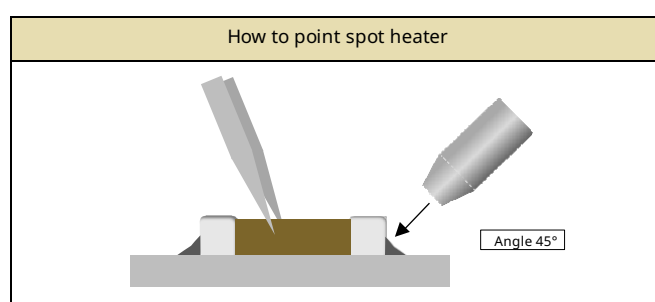


Soldering Method

- 1) Ceramic is easily damaged by rapid heating or cooling. If some heat shock is unavoidable, preheat enough to limit the temperature difference (Delta T) to within 150 degree Celsius.
- 2) The product size 1.6×0.8mm to 3.2×1.6mm can be used in reflow and wave soldering, and the product size of bigger than 3.2×1.6mm, or smaller than 1.6×0.8mm can be used in reflow.
Circuit shortage and smoking can be created by using capacitors which are used neglecting the above caution.
- 3) Please see our recommended soldering conditions.
- 4) In case of using Sn-Zn Solder, please contact us in advance.
- 5) The following condition is recommended for spot heater application.

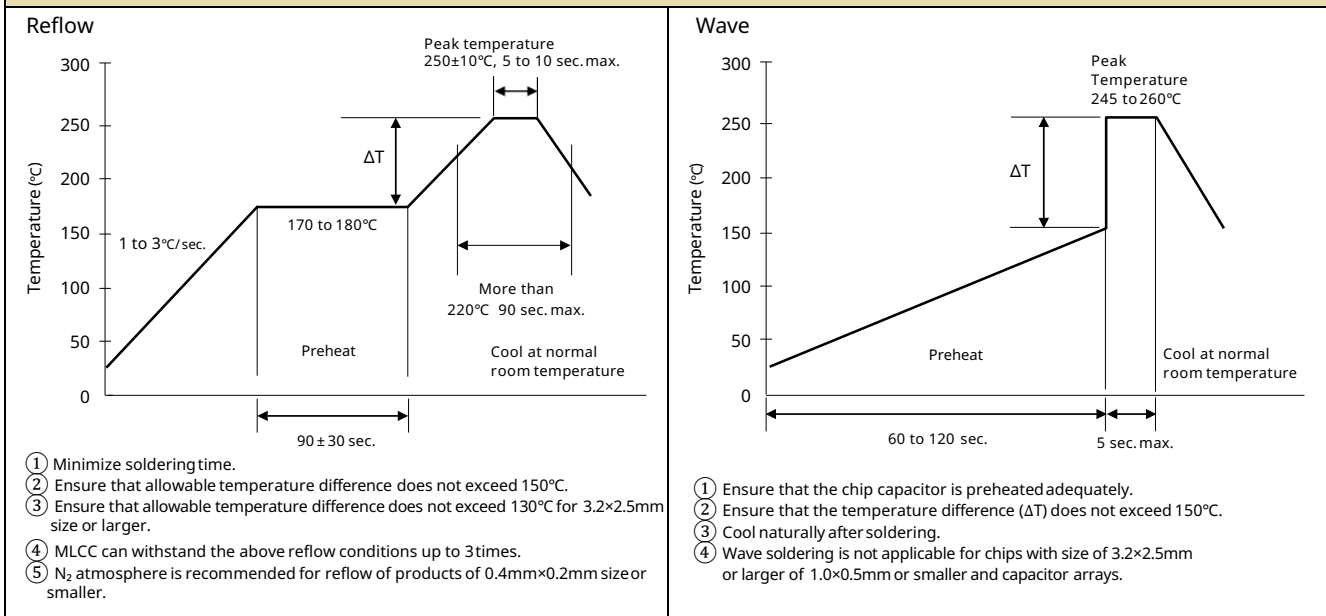
Recommended spot heater condition

Item	Condition
Distance	5mm min.
Angle	45°
Projection Temp.	400°C max.
Flow rate	Set at the minimum
Nozzle diameter	2φ to 4φ
Application time	10 sec. max. (1206 and smaller) 30 sec. max. (1210 and larger)



Multilayer Ceramic Chip Capacitors Surface Mounting Information

Recommended Temperature Profile (Sn-3.0Ag-0.5Cu)



Precautions after board mounting

Excessive board warpage during the mounting of chip capacitors, or when breaking the board after mounting, may cause cracks or fractures in the capacitors. Please ensure that board warpage is kept to an absolute minimum.

Resin Mold

- 1) If a large amount of resin is used for molding the chip, cracks may occur due to contraction stress during curing. To avoid such cracks, use a low shrinkage resin.
- 2) The insulation resistance of the chip will degrade due to moisture absorption. Use a low moisture absorption resin.
- 3) Check carefully that the resin does not generate a decomposition gas or reaction gas during the curing process or during normal storage. Such gases may crack the chip capacitor or damage the device itself.
- 4) If used in an environment containing sulfur compounds, silver migration may occur, leading to insulation degradation.

Multilayer Ceramic Chip Capacitors Precautions

Circuit Design

1. Once application and assembly environments have been checked, the capacitor may be used in conformance with the rating and performance which are provided in both the catalog and the specifications. Use exceeding that which is specified may result in inferior performance or cause a short, open, smoking, or flaming to occur, etc.
2. Please consult the manufacturer in advance when the capacitor is used in devices such as: devices which deal with human life, i.e. medical devices; devices which are highly public orientated; and devices which demand a high standard of liability.
Accident or malfunction of devices such as medical devices, space equipment and devices having to do with atomic power could generate grave consequence with respect to human lives or, possibly, a portion of the public. Capacitors used in these devices may require high reliability design different from that of general purpose capacitors.
3. Please use the capacitors in conformance with the operating temperature provided in both the catalog and the specifications.
Be especially cautious not to exceed the maximum temperature. In the situation the maximum temperature set forth in both the catalog and specifications is exceeded, the capacitor's insulation resistance may deteriorate, power may suddenly surge and short-circuit may occur.
The capacitor has a loss, and may self-heat due to equivalent series resistance when alternating electric current is passed therethrough. As this effect becomes especially pronounced in high frequency circuits, please exercise caution.
When using the capacitor in a (self-heating) circuit, please make sure the surface of the capacitor remains under the maximum temperature for usage. Also, please make certain temperature rises remain below 20°C.
4. Please keep voltage under the rated voltage which is applied to the capacitor. Also, please make certain the peak voltage remains below the rated voltage when AC voltage is super-imposed to the DC voltage.
In the situation where AC or pulse voltage is employed, ensure average peak voltage does not exceed the rated voltage.
Exceeding the rated voltage provided in both catalog and specifications may lead to defective withstanding voltage or, in worst case situations, may cause the capacitor to smoke or flame.
5. When the capacitor is to be employed in a circuit in which there is continuous application of a high frequency voltage or a steep pulse voltage, even though it is within the rated voltage, please inquire to the manufacturer. In the situation the capacitor is to be employed using a high frequency AC voltage or a extremely fast rising pulse voltage, even though it is within the rated voltage, it is possible capacitor reliability will deteriorate.
6. It is a common phenomenon of high-dielectric products to have a deteriorated amount of static electricity due to the application of DC voltage.
Due caution is necessary as the degree of deterioration varies depending on the quality of capacitor materials, capacity, as well as the load voltage at the time of operation.
7. Do not use the capacitor in an environment where it might easily exceed the respective provisions concerning shock and vibration specified in the catalog and specifications.
In addition, it is a common piezo phenomenon of high dielectric products to have some voltage due to vibration or to have noise due to voltage change. Please contact sales in such case.
8. If the electrostatic capacity value of the delivered capacitor is within the specified tolerance, please consider this when designing the respective product in order that the assembled product function appropriately.
9. Please contact us upon using conductive adhesives.

Storage

Please note the following regarding the storage of delivered products.

1. Set the storage temperature to + 5 to + 40 °C and humidity to 20 ~ 70% RH. Other meteorological conditions are in accordance with classification 1 K2 of JIS C 60721 -3 -1.
2. Store in a place where corrosive gas (H₂S, SO₂, NO₂, Cl₂, etc.) does not exist in the atmosphere. Also, avoid exposure to salty moisture. In either case, this may cause oxidation corrosion of the terminal electrode, reducing solderability.

If you store the above delivered products according to the conditions listed above, it will satisfy the solderability standard for 6 months from the shipping date.

Safety application guideline and detailed information of electrical properties are also provided in kyocera web site;

URL: <https://ele.kyocera.com/en/product/capacitor/>



Multilayer Ceramic Chip Capacitors

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